Technical Note: block-VIEM.jl

Volume Integral Equation Method with SWG/half-SWG Basis,
Duffy Singular Integration, and AIM-FFT Acceleration
for Light Scattering by Arbitrary Dielectric Particles

N. Moteki

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Contents

1 Introduction

`block-VIEM.jl` is a Julia implementation of the volume integral equation method (VIEM) for electromagnetic scattering by arbitrarily shaped, high-contrast dielectric particles. It is designed as a higher-accuracy successor to the discrete dipole approximation (DDA) code `block-DDA_Py` [11], targeting scattering problems where DDA does not converge or provides insufficient accuracy — for example, particles with high refractive-index contrast (e.g. plasmonic metals), extreme aspect ratios, or surface roughness features smaller than the DDA lattice spacing.

The key features of `block-VIEM.jl` are:

- **Higher accuracy per degree of freedom than DDA.** Tetrahedral mesh discretization with SWG (half-SWG) basis functions [1] and Duffy-transform singular integration achieves 10–12× better accuracy per DOF than cubic-lattice DDA on equivalent problems.
- **AIM/FFT-accelerated block-Krylov solver.** The Adaptive Integral Method (AIM) [5] computes matrix–vector products in $O(N \log N)$ via FFT. Block BiCGSTAB and Block GMRES solve for all particle orientations simultaneously.
- **Multi-orientation batch solve with spheroid mode.** Deterministic Euler angle grids on SO(3) identical to `block-DDA_Py`. For spheroids, only N_β orientations are solved; the full grid is filled analytically.
- **Birefringent (anisotropic) materials.** Diagonal permittivity tensor $\epsilon_p = \mathrm{diag}(\epsilon_x, \epsilon_y, \epsilon_z)$.
- **Gaussian Random Ellipsoid (GRE) shape model with automatic mesh generation.** Parametric irregular shapes matching `block-DDA_Py`, with adaptive tetrahedral meshing via Gmsh.
- **CAS-v2 observables and block-DDA_Py-compatible I/O.** Forward/backward complex scattering amplitudes (PCAS, OCBS), cross sections, and HDF5 parameter-sweep output in the same schema as `block-DDA_Py`.

The two codes share the same physical input conventions, HDF5 output schema, and CAS-v2 observable definitions, so downstream analysis pipelines work with either without modification.

Scope of this note. This document describes the theoretical background and algorithms implemented in `block-VIEM.jl`, covering:

1. the EFVIE-D formulation with SWG and half-SWG basis functions (Sec. 2),
2. weakening the $1/R^3$ singularity to $1/R$ via integration by parts, including the boundary-face surface-correction terms K^B, K^C, K^D required to restore charge neutrality (Sec. 3),
3. evaluation of the remaining $1/R$ singular and near-singular volume integrals via the generalized Duffy transform (Sec. 4),
4. FFT-accelerated matrix–vector products through the Adaptive Integral Method (AIM) with moment matching and precorrection (Sec. 5),
5. the direct and Krylov iterative solvers and the multi-orientation batch solve (Sec. 6),
6. the CAS-v2 compatible scattering observables computed from the solution (Sec. 7),
7. validation against Mie theory and the oblate-spheroid symmetry identity (Sec. 8),
8. the HDF5 output format for spheroid parameter sweeps, bit compatible with the `block-DDA_Py` schema consumed by `build_spheroid_lut.py` (Sec. 9).

Notation and conventions. Throughout this note the electromagnetic equations are written in the physics (BH83 [10]) time convention $\mathrm{e}^{-i\omega t}$, so that an outgoing scalar Helmholtz Green's function carries the factor $\mathrm{e}^{+ik_0 R}/(4\pi R)$. Absorbing materials have $\mathrm{Im}(\epsilon_p) > 0$. Lengths are expressed in micrometres (µm). Vectors are set in bold italic ($\boldsymbol{E}$, $\boldsymbol{k}$, $\boldsymbol{r}$), matrices and second-rank tensors in upright bold ($\mathbf{A}$, $\mathbf{G}$), and unit vectors carry a hat ($\hat{\boldsymbol{e}}$, $\hat{\boldsymbol{n}}$).

2 VIEM Formulation

2.1 EFVIE-D strong form

Consider a non-magnetic ($\mu = \mu_0$) inhomogeneous dielectric particle occupying the volume Ω with complex permittivity $\epsilon_p(\boldsymbol{r})$, embedded in a homogeneous background of real permittivity ϵ_{bg}. The wavenumber in the background is

$$k_0 = \omega\,\sqrt{\mu_0\,\epsilon_{\mathrm{bg}}} = \frac{2\pi\,m_m}{\lambda_0}, \tag{1}$$

where $m_m = \sqrt{\epsilon_{\mathrm{bg}}/\epsilon_0}$ is the refractive index of the background medium.

We use the electric flux volume integral equation in the *D-form* [1]: the unknown field is the electric flux density $\boldsymbol{D}(\boldsymbol{r}) = \epsilon_p(\boldsymbol{r})\,\boldsymbol{E}(\boldsymbol{r})$, whose normal component is continuous across material interfaces, so the scheme places all dielectric jumps inside the basis functions and not in the expansion coefficients. Define the dielectric contrast

$$\kappa(\boldsymbol{r}) = \frac{\epsilon_p(\boldsymbol{r}) - \epsilon_{\mathrm{bg}}}{\epsilon_p(\boldsymbol{r})}. \tag{2}$$

The equivalent volume polarisation current is $\boldsymbol{J}_v = -i\omega\,\kappa\,\boldsymbol{D}$, and the EFVIE-D strong form reads [1, 2]

$$\boldsymbol{E}_{\mathrm{inc}}(\boldsymbol{r}) = \frac{\boldsymbol{D}(\boldsymbol{r})}{\epsilon_p(\boldsymbol{r})} - \frac{1}{\epsilon_{\mathrm{bg}}}\big(k_0^2 + \boldsymbol{\nabla}\boldsymbol{\nabla}\cdot\big)\int_\Omega \kappa(\boldsymbol{r}')\,\boldsymbol{D}(\boldsymbol{r}')\,G(\boldsymbol{r},\boldsymbol{r}')\,\mathrm{d}^3 r', \tag{3}$$

where $G(\boldsymbol{r},\boldsymbol{r}')$ is the free-space scalar Helmholtz Green's function

$$G(\boldsymbol{r},\boldsymbol{r}') = \frac{\mathrm{e}^{+ik_0 R}}{4\pi R}, \qquad R = \big|\boldsymbol{r} - \boldsymbol{r}'\big|. \tag{4}$$

2.2 Incident plane wave

The incident field is a monochromatic plane wave

$$\boldsymbol{E}_{\mathrm{inc}}(\boldsymbol{r}) = \boldsymbol{E}_0\,\exp\!\big(+ik_0\,\hat{\boldsymbol{k}}_{\mathrm{inc}}\cdot\boldsymbol{r}\big), \tag{5}$$

with $\hat{\boldsymbol{k}}_{\mathrm{inc}}$ the unit propagation direction and $\boldsymbol{E}_0$ a complex polarisation vector perpendicular to $\hat{\boldsymbol{k}}_{\mathrm{inc}}$. For CAS-v2 observables the incident polarisation is left-handed circular (matching `block-DDA_Py` [11]):

$$\boldsymbol{E}_0 = \frac{1}{\sqrt{2}}\,\hat{\boldsymbol{e}}_{\theta,\mathrm{inc}} + \frac{i}{\sqrt{2}}\,\hat{\boldsymbol{e}}_{\phi,\mathrm{inc}}, \tag{6}$$

where $\hat{\boldsymbol{e}}_{\theta,\mathrm{inc}}$, $\hat{\boldsymbol{e}}_{\phi,\mathrm{inc}}$ are the polar and azimuthal unit vectors of the incident direction in the particle frame. Particle orientation is parameterised by intrinsic ZYZ Euler angles (α,β,γ), following `scipy.spatial.transform.Rotation` and `block-DDA_Py`. We denote by

$$\mathbf{R}(\alpha,\beta,\gamma) \;=\; \mathbf{R}_z(\alpha)\,\mathbf{R}_y(\beta)\,\mathbf{R}_z(\gamma) \tag{7}$$

the active rotation that maps particle-frame coordinates of a physical vector to its lab-frame coordinates; equivalently, (7) is the fixed-frame composition obtained by applying the three extrinsic rotations $\mathbf{R}_z(\gamma)$, $\mathbf{R}_y(\beta)$, $\mathbf{R}_z(\alpha)$ in that order about the lab axes. The laboratory-frame triad $\{\hat{\boldsymbol{u}}_{\mathrm{inc},L}, \hat{\boldsymbol{e}}_{\theta,L}, \hat{\boldsymbol{e}}_{\phi,L}\}$ is therefore expressed in the particle frame by applying $\mathbf{R}(\alpha,\beta,\gamma)^\top$ ($= \mathbf{R}(\alpha,\beta,\gamma)^{-1}$ since $\mathbf{R}$ is orthogonal).

2.3 SWG and half-SWG basis functions

The particle volume Ω is discretised into a conforming tetrahedral mesh $\mathcal{T}_h$ with N_{tet} elements. The mesh is generated by Gmsh with a target characteristic edge length ℓ_c, which controls the maximum edge length of every tetrahedron; smaller ℓ_c yields a finer mesh with more elements and degrees of freedom. On each internal face S_n shared by two tetrahedra T_n^+ and T_n^-, an SWG basis function [1, Eqs. (10), (12)] is defined by

$$\boldsymbol{f}_n(\boldsymbol{r}) = \begin{cases} \dfrac{a_n}{3\,V_n^+}\left(\boldsymbol{r} - \boldsymbol{p}_n^+\right), & \boldsymbol{r} \in T_n^+, \\ -\dfrac{a_n}{3\,V_n^-}\left(\boldsymbol{r} - \boldsymbol{p}_n^-\right), & \boldsymbol{r} \in T_n^-, \\ \boldsymbol{0}, & \text{otherwise}, \end{cases} \tag{8}$$

where a_n is the face area, $V_n^\pm$ the volumes of the two supporting tetrahedra, and $\boldsymbol{p}_n^\pm$ the free vertex (the vertex of $T_n^\pm$ not on S_n). The normalisation in (8) enforces $\boldsymbol{f}_n \cdot \hat{\boldsymbol{n}} = 1$ on S_n (oriented from T_n^+ to T_n^-) and zero on the other five faces of $T_n^+ \cup T_n^-$. The divergence is piecewise constant,

$$\boldsymbol{\nabla} \cdot \boldsymbol{f}_n(\boldsymbol{r}) = \begin{cases} +\dfrac{a_n}{V_n^+}, & \boldsymbol{r} \in T_n^+, \\ -\dfrac{a_n}{V_n^-}, & \boldsymbol{r} \in T_n^-, \\ 0, & \text{otherwise}. \end{cases} \tag{9}$$

A naive SWG expansion that uses only internal faces implicitly imposes $\boldsymbol{D} \cdot \hat{\boldsymbol{n}} = 0$ on the particle boundary $\partial\Omega$, leaving the surface polarisation charge unrepresented and producing systematic cross-section errors of order 10–30% on Mie tests. To restore charge neutrality, the original [1] formulation adds one half-SWG basis function per boundary face. On a boundary face S_n with a single supporting tetrahedron T_n^+, the half-SWG function is the first branch of (8) alone:

$$\boldsymbol{f}_n^{\text{half}}(\boldsymbol{r}) = \frac{a_n}{3\,V_n^+}\left(\boldsymbol{r} - \boldsymbol{p}_n^+\right), \qquad \boldsymbol{r} \in T_n^+, \tag{10}$$

with $\boldsymbol{f}_n \cdot \hat{\boldsymbol{n}} = 1$ on S_n and zero on the other three faces of T_n^+. For a half-SWG function the normal flux does *not* vanish on $\partial\Omega$, so it carries an induced surface polarisation charge $\sigma = \kappa\,\boldsymbol{D} \cdot \hat{\boldsymbol{n}}$ whose integral over S_n exactly cancels the bulk bound charge of the same basis (Sec. 3.3). The electric flux density is expanded as

$$\boldsymbol{D}(\boldsymbol{r}) \approx \sum_{n=1}^{N_{\text{dof}}} D_n\,\boldsymbol{f}_n(\boldsymbol{r}), \qquad N_{\text{dof}} = N_{\text{int. faces}} + N_{\text{bdy. faces}}. \tag{11}$$

2.4 Galerkin weak form

Testing (3) with a basis function $\boldsymbol{f}_m$ and substituting (11) yields the $N_{\text{dof}} \times N_{\text{dof}}$ linear system

$$\sum_{n=1}^{N_{\text{dof}}} Z_{mn}\,D_n = b_m, \qquad b_m = \int_\Omega \boldsymbol{f}_m(\boldsymbol{r}) \cdot \boldsymbol{E}_{\text{inc}}(\boldsymbol{r})\,\mathrm{d}^3 r, \tag{12}$$

with the Galerkin impedance matrix element

$$\begin{aligned} Z_{mn} = {} & \int_\Omega \frac{\boldsymbol{f}_m(\boldsymbol{r}) \cdot \boldsymbol{f}_n(\boldsymbol{r})}{\epsilon_p(\boldsymbol{r})}\,\mathrm{d}^3 r \\ & - \frac{1}{\epsilon_{\text{bg}}} \int_\Omega \boldsymbol{f}_m(\boldsymbol{r}) \cdot \Big[\left(k_0^2 + \boldsymbol{\nabla}\boldsymbol{\nabla}\cdot\right) \int_\Omega \kappa(\boldsymbol{r}')\,\boldsymbol{f}_n(\boldsymbol{r}')\,G(\boldsymbol{r}, \boldsymbol{r}')\,\mathrm{d}^3 r'\Big]\mathrm{d}^3 r. \end{aligned} \tag{13}$$

For a homogeneous particle, κ and ϵ_p are constants and factor out of the inner integral, giving the compact decomposition

$$Z_{mn} = \frac{1}{\epsilon_p} M_{mn} - \frac{\kappa}{\epsilon_{\text{bg}}} K_{mn}, \tag{14}$$

where $M_{mn} = \int \boldsymbol{f}_m \cdot \boldsymbol{f}_n \, \mathrm{d}^3 r$ is the mass matrix and K_{mn} is the radiation kernel defined below.

3 Weakening the Singularity via Integration by Parts

As written in (13), the radiation kernel $(k_0^2 + \boldsymbol{\nabla}\boldsymbol{\nabla}\cdot)G$ contains a hyper-singular term $1/R^3$ coming from $\boldsymbol{\nabla}\boldsymbol{\nabla} \cdot G$. Moving the two derivatives off G onto $\boldsymbol{f}_m$ and $\boldsymbol{f}_n$ by successive integrations by parts reduces the singularity to the weakly singular $1/R$ kernel that Duffy quadrature can handle. Because SWG and half-SWG functions are only piecewise smooth and only half-SWG functions have nonzero $\boldsymbol{f}_n \cdot \hat{\boldsymbol{n}}$ on $\partial\Omega$, the by-parts procedure generates three boundary-face contributions in addition to the standard bulk term.

3.1 Bulk–bulk term K^A

Apply $\boldsymbol{\nabla}\boldsymbol{\nabla}\cdot$ to the inner integral and move the inner divergence onto $\boldsymbol{f}_n$ using the divergence theorem; do the same for the outer divergence and $\boldsymbol{f}_m$. For internal SWG basis functions (vanishing normal flux on $\partial\Omega$) the boundary terms are identically zero and the resulting bulk kernel is

$$K^A_{mn} = \int_\Omega \int_\Omega \left[k_0^2 \, \boldsymbol{f}_m(\boldsymbol{r}) \cdot \boldsymbol{f}_n(\boldsymbol{r}') - (\boldsymbol{\nabla} \cdot \boldsymbol{f}_m)(\boldsymbol{r}) \, (\boldsymbol{\nabla} \cdot' \boldsymbol{f}_n)(\boldsymbol{r}') \right] G(\boldsymbol{r}, \boldsymbol{r}') \, \mathrm{d}^3 r' \, \mathrm{d}^3 r. \tag{15}$$

The integrand now contains only the weakly singular $1/R$ kernel and is evaluated tetrahedron pair by tetrahedron pair.

3.2 Surface correction terms K^B, K^C, K^D

When the source basis $\boldsymbol{f}_n$ is a half-SWG function attached to a boundary face S_n, the inner integration by parts leaves a surface term from the boundary $\partial\Omega$:

$$\Phi_n(\boldsymbol{r}) \equiv \boldsymbol{\nabla} \cdot \int_\Omega G(\boldsymbol{r}, \boldsymbol{r}') \, \boldsymbol{f}_n(\boldsymbol{r}') \, \mathrm{d}^3 r' = \int_\Omega G \, \boldsymbol{\nabla} \cdot' \boldsymbol{f}_n \, \mathrm{d}^3 r' \; - \oint_{\partial\Omega} G(\boldsymbol{r}, \boldsymbol{r}') \left(\boldsymbol{f}_n \cdot \hat{\boldsymbol{n}}' \right) \mathrm{d}^2 r'. \tag{16}$$

Similarly, the outer integration by parts against $\boldsymbol{f}_m$ produces a surface term if $\boldsymbol{f}_m$ is a half-SWG function on a boundary face S_m. The complete kernel is

$$K_{mn} = K^A_{mn} + K^B_{mn} + K^C_{mn} + K^D_{mn}, \tag{17}$$

with the three boundary contributions

$$K^B_{mn} = \int_\Omega \oint_{\partial\Omega} (\boldsymbol{\nabla} \cdot \boldsymbol{f}_m)(\boldsymbol{r}) \left(\boldsymbol{f}_n \cdot \hat{\boldsymbol{n}}' \right) G(\boldsymbol{r}, \boldsymbol{r}') \, \mathrm{d}^2 r' \, \mathrm{d}^3 r, \tag{18}$$

$$K^C_{mn} = \oint_{\partial\Omega} \int_\Omega \left(\boldsymbol{f}_m \cdot \hat{\boldsymbol{n}} \right) (\boldsymbol{\nabla} \cdot' \boldsymbol{f}_n)(\boldsymbol{r}') \, G(\boldsymbol{r}, \boldsymbol{r}') \, \mathrm{d}^3 r' \, \mathrm{d}^2 r, \tag{19}$$

$$K^D_{mn} = -\oint_{\partial\Omega} \oint_{\partial\Omega} \left(\boldsymbol{f}_m \cdot \hat{\boldsymbol{n}} \right) \left(\boldsymbol{f}_n \cdot \hat{\boldsymbol{n}}' \right) G(\boldsymbol{r}, \boldsymbol{r}') \, \mathrm{d}^2 r' \, \mathrm{d}^2 r. \tag{20}$$

By the half-SWG property $\boldsymbol{f}_n \cdot \hat{\boldsymbol{n}} = 1$ on S_n and zero on the remaining boundary faces, the surface integrals in (18)–(20) collapse to integrals over the specific faces S_m and S_n: K^B_{mn} is non-zero only when n is a boundary face, K^C_{mn} only when m is a boundary face, and K^D_{mn} only when both m and n are boundary faces.

3.3 Charge neutrality

The induced bound charge associated with a single half-SWG basis on a boundary face S_n is the sum of a bulk contribution $\rho_{\mathrm{vol}} = -\kappa\, \boldsymbol{\nabla} \cdot \boldsymbol{D}$ and a surface contribution $\sigma_{\mathrm{surf}} = \kappa\, \boldsymbol{D} \cdot \hat{\boldsymbol{n}}$. Using (9) and the unit-flux property on S_n,

$$Q_{\mathrm{vol}} = -\int_{T_n^+} \kappa\, \boldsymbol{\nabla} \cdot \boldsymbol{f}_n \,\mathrm{d}^3 r = -\kappa \frac{a_n}{V_n^+} \cdot V_n^+ = -\kappa\, a_n, \tag{21}$$

$$Q_{\mathrm{surf}} = \int_{S_n} \kappa\, (\boldsymbol{f}_n \cdot \hat{\boldsymbol{n}}) \,\mathrm{d}^2 r = \kappa\, a_n. \tag{22}$$

The sum $Q_{\mathrm{vol}} + Q_{\mathrm{surf}} = 0$ holds exactly *only* when the surface contributions K^B, K^C, K^D are retained. Dropping them, as an internal-only SWG expansion does, leaves the particle with a spurious net bound charge $-\kappa\, a_n$ per boundary face and produces cross-section errors of order 10–30% on Mie tests. With the full decomposition (17), block-VIEM.jl reproduces the Mie sphere absorption cross section to $\sim 0.2\%$ at $\ell_c = 0.5$ (589 DoFs), an order-of-magnitude improvement per DoF over cubic-lattice DDA.

3.4 Mass matrix

The first term in (14) is the sparse mass matrix

$$M_{mn} = \int_{T_m \cap T_n} \boldsymbol{f}_m(\boldsymbol{r}) \cdot \boldsymbol{f}_n(\boldsymbol{r}) \,\mathrm{d}^3 r. \tag{23}$$

The integrand is a degree-2 polynomial in each tetrahedron, so the 5-point Gauss rule on the tetrahedron (degree 3 exact) evaluates M_{mn} without quadrature error.

4 Singular and Near-Singular Volume Integration

4.1 Generalised vertex Duffy transform

The weakly singular kernel $1/R$ in (15)–(20) remains integrable but standard Gauss quadrature fails when the source and test tetrahedra share a vertex, edge, or face, or when the outer Gauss point approaches a face of the inner element. Following Mousavi–Sukumar [4], all such cases are handled uniformly by a *vertex-only* Duffy transform applied to sub-tetrahedra obtained by projecting the singular point onto the inner element.

For $\alpha = 1$ (a $1/R$ singularity at a single vertex), the optimised transform from the unit cube $\boldsymbol{u} = (u_1, u_2, u_3) \in [0, 1]^3$ to barycentric coordinates $(\lambda_1, \lambda_2, \lambda_3, \lambda_4)$ with $\lambda_1 = 1$ at the singular vertex is [4, Eq. (1) with $\beta = 1$]

$$\lambda_1 = 1 - u_1, \quad \lambda_2 = u_1(1 - u_2), \quad \lambda_3 = u_1 u_2 (1 - u_3), \quad \lambda_4 = u_1 u_2 u_3. \tag{24}$$

The face $u_1 = 0$ of the cube collapses entirely to the singular vertex, and $R \propto u_1$ to leading order along any ray departing from it. The total Jacobian combining (24) with the affine map from reference to physical coordinates is

$$J_D(\boldsymbol{u}) = 6\, |V_{\mathrm{tet}}|\, u_1^2\, u_2. \tag{25}$$

The factor u_1^2 exactly cancels the $1/R$ singularity, leaving a smooth integrand on the unit cube that is integrated by a tensor-product Gauss–Legendre rule. For the K^A kernel (15) the recommended order is $5 \times 5 \times 5$ (125 points) for self and face-adjacent pairs, and a $4 \times 4 \times 4$ rule for edge- and vertex-adjacent pairs; a 5-point Gauss rule on the source tetrahedron is sufficient for all pairs that are not near the observation point.

4.2 Pair classification

Given an outer quadrature point $\boldsymbol{r}$ inside the test tetrahedron T_m, the source element T_n is classified and integrated according to the minimum distance:

- *self* ($T_n = T_m$): the outer Gauss point is the singular vertex. The source tetrahedron is subdivided into four sub-tetrahedra with the outer Gauss point as a common vertex, and each is integrated by the Duffy transform (24).

- *face-/edge-/vertex-adjacent*: the singular point is the nearest vertex, the nearest point on the shared edge, or the shared face. Sub-tetrahedra are generated as above around that singular point and integrated by the same Duffy rule.

- *far* (R larger than a chosen threshold): the inner integral is evaluated with a plain tetrahedron Gauss rule (5 points). Pairs further than the AIM near-field radius are never integrated directly (Sec. 5).

4.3 Surface integrals for K^B, K^C, K^D

The surface contributions in Sec. 3.2 require the integration of G over a triangle, optionally combined with an inner tetrahedron integral (K^B, K^C) or an outer triangle integral (K^D). We adopt a two-dimensional version of the Duffy transform on the reference triangle: from $(u_1, u_2) \in [0, 1]^2$ to barycentric coordinates (η_1, η_2, η_3),

$$\eta_1 = 1 - u_1, \quad \eta_2 = u_1(1 - u_2), \quad \eta_3 = u_1 u_2, \tag{26}$$

with Jacobian $J_{D,2}(\boldsymbol{u}) = 2|A_{\mathrm{tri}}|\, u_1$. The linear factor u_1 cancels the $1/R$ singularity at the vertex $\eta_1 = 1$, and the face $u_1 = 0$ collapses to that vertex just as in the three-dimensional case. For the K^D self/edge/vertex branches, the outer triangle is first subdivided into sub-triangles at the singular feature and the 2D Duffy rule is applied to each.

The near-singular sub-cases of K^B and K^C (for example, the outer tetrahedron T_m having a face coincident with the inner boundary face S_n) are handled by projecting the outer Gauss point onto the inner triangle and applying the 2D Duffy rule around that projection.

5 FFT-Accelerated MVP: Adaptive Integral Method

Direct computation and storage of the dense impedance matrix Z scales as $\mathcal{O}(N_{\mathrm{dof}}^2)$ in memory and $\mathcal{O}(N_{\mathrm{dof}}^3)$ in operations for direct factorisation, quickly becoming prohibitive. The Adaptive Integral Method (AIM) [3] accelerates the matrix–vector product by representing the far-field part of the kernel on an auxiliary cubic grid and evaluating the resulting block-Toeplitz sum with a three-dimensional FFT, mirroring the Goodman–Draine–Flatau construction [9] used by `block-DDA_Py`.

5.1 Auxiliary grid and stencil

A Cartesian grid of spacing h_g is overlaid on the bounding box of the particle. Let $\boldsymbol{g}_p$ denote the position of grid point p. Each basis function $\boldsymbol{f}_n$ is associated with a stencil $\mathcal{S}_n$ of $M_{\mathrm{st}} = (n_{\mathrm{st}}+1)^3$ nearby grid points surrounding its centroid. For a moment matching order n_{st}, the stencil is a cube with $n_{\mathrm{st}} + 1$ grid points on each side.

5.2 Moment matching

For each basis function $\boldsymbol{f}_n$ and each stencil point $p \in \mathcal{S}_n$ we construct a projection weight $W_{np} \in \mathbb{C}^3$ so that the multipole moments of the point source at p match those of the continuous source $\boldsymbol{f}_n$ up to order n_{st}. The moments are defined with respect to the basis centroid $\boldsymbol{c}_n$:

$$\mu_n^{\boldsymbol{\alpha}} = \int_\Omega \boldsymbol{f}_n(\boldsymbol{r})\,(\boldsymbol{r}-\boldsymbol{c}_n)^{\boldsymbol{\alpha}}\,\mathrm{d}^3 r, \qquad |\boldsymbol{\alpha}| = \alpha_x+\alpha_y+\alpha_z \le n_{\mathrm{st}}, \tag{27}$$

and the weights $\{W_{np}\}_{p\in\mathcal{S}_n}$ are defined by the linear constraint

$$\sum_{p\in\mathcal{S}_n} W_{np}\,(\boldsymbol{g}_p-\boldsymbol{c}_n)^{\boldsymbol{\alpha}} = \mu_n^{\boldsymbol{\alpha}}, \qquad \forall\,|\boldsymbol{\alpha}|\le n_{\mathrm{st}}. \tag{28}$$

A companion set of weights matches the divergence moments $\int(\boldsymbol{\nabla}\cdot\boldsymbol{f}_n)(\boldsymbol{r}-\boldsymbol{c}_n)^{\boldsymbol{\alpha}}\,\mathrm{d}^3 r$ for the scalar potential part of the kernel. The linear system (28) is small (M_{st} unknowns per basis function, $\binom{n_{\mathrm{st}}+3}{3}$ equations) and is solved once, during matrix setup, per basis function.

5.3 Far-field product via FFT

With the moment matching complete, the far-field part of the matrix–vector product $\boldsymbol{y}^{\mathrm{far}} = K^{\mathrm{far}}\,\boldsymbol{x}$ is evaluated in three steps:

1. *Projection.* For each basis function n and each stencil point p, accumulate $q_p = \sum_n W_{np}\,x_n$ onto the grid.
2. *Convolution.* Compute $Q_p = \sum_{p'} G(\boldsymbol{g}_p-\boldsymbol{g}_{p'})\,q_{p'}$ via a three-dimensional FFT of q, pointwise multiplication by the precomputed FFT of G on the doubled grid, and an inverse FFT.
3. *Interpolation.* For each test function m, $y_m^{\mathrm{far}} = \sum_{p\in\mathcal{S}_m} W_{mp}^{\top}\,Q_p$.

Because G depends only on the grid-point separation, its convolution matrix is block-Toeplitz and is computed with the 3D FFT of the same construction used in `block-DDA_Py` [9]. The cost is $\mathcal{O}(N_g\log N_g)$ per MVP, where N_g is the number of grid points. Memory for the grid Green function is also $\mathcal{O}(N_g)$ on the doubled domain.

5.4 Precorrection of near-field pairs

The FFT sum above is accurate only in the far field; for pairs with $\|\boldsymbol{c}_m-\boldsymbol{c}_n\|\le r_{\mathrm{near}}$ it is replaced by the exact integrals of Sec. 4. The construction is a *difference* of two quantities: for each near pair (m,n) we compute the exact K_{mn} and subtract from it the contribution that the FFT would have produced through the grid projection, storing the difference in a sparse precorrection matrix K^{pc}. The complete MVP is

$$\boldsymbol{y} = \big(D_M - \tfrac{\kappa}{\epsilon_{\mathrm{bg}}}(K^{\mathrm{far}}+K^{\mathrm{pc}})\big)\,\boldsymbol{x}, \tag{29}$$

where D_M is the sparse diagonal mass contribution M/ϵ_p and K^{pc} has $\mathcal{O}(N_{\mathrm{dof}})$ non-zero entries. The default near radius is $r_{\mathrm{near}} = 2\times(\text{stencil extent})$, verified in Phase-3 benchmarks to produce $< 0.5\%$ relative error between the AIM-accelerated and the dense matrix–vector products.

5.5 Surface-moment extension for K^B, K^C, K^D

The projection of Sec. 5.2 accelerates only the bulk–bulk kernel K^A of Eq. (15). The three boundary-face contributions K^B, K^C, K^D of Eqs. (18)–(20) are, in the Phase-1a release, stored as a separate sparse matrix `half_swg_extra` of dimension $N_{\mathrm{bnd}}\times N_{\mathrm{dof}}$, which for the iron-oxide doublet geometry dominates total live memory at 81–86% (measured at R/5, R/6 mesh refinement). This subsection extends the moment-matching projection to the boundary-face supports so that K^B, K^C, K^D share the same three-dimensional FFT convolution as K^A, reducing `half_swg_extra` to a sparse near-field correction block whose storage scales as $\mathcal{O}(N_{\mathrm{bnd}})$ rather than $\mathcal{O}(N_{\mathrm{bnd}}N_{\mathrm{dof}})$. The construction follows Anastassiu *et al.* [6], which supplies the mathematical rigour ("thin-sheet constraint") omitted from the original AIM presentation of Bleszynski *et al.* [5].

Scalar surface density on a volumetric grid

Inspection of Eqs. (18)–(20) shows that the three boundary kernels couple to a *single* scalar surface density

$$\sigma_n(\boldsymbol{r}) \equiv \boldsymbol{f}_n(\boldsymbol{r})\cdot\hat{\boldsymbol{n}}(\boldsymbol{r}), \qquad \boldsymbol{r} \in S_n, \tag{30}$$

supported on the single boundary triangle S_n of basis n; by the half-SWG unit-flux property, $\sigma_n = 1$ on S_n and vanishes elsewhere on $\partial\Omega$. Because the divergence theorem of Sec. 3.2 has already contracted the vector basis $\boldsymbol{f}_n$ against the outward normal, the surface AIM projection requires neither Cartesian components nor a surface divergence of $\boldsymbol{f}_n$: only the scalar σ_n appears. This is a substantial simplification over the four-channel RWG projection of Anastassiu [6], where no preceding integration by parts is performed.

The triangle S_n is a two-dimensional manifold embedded in the three-dimensional grid $\{\boldsymbol{g}_p\}$. A naive surface moment $\int_{S_n} \sigma_n\,(\boldsymbol{r}-\boldsymbol{c}_n)^{\boldsymbol{\alpha}}\,\mathrm{d}^2r$ determines the grid projection only for multi-indices $\boldsymbol{\alpha}$ with zero normal component; moments with $\alpha_\perp \geq 1$ (where $\perp$ denotes the local direction along $\hat{\boldsymbol{n}}$) remain unconstrained, and the projection is consequently rank-deficient on a volumetric grid. Following Anastassiu [6], we *thicken* σ_n to a volumetric distribution by a normal delta,

$$\tilde{\sigma}_n(\boldsymbol{r}) \equiv \sigma_n(\boldsymbol{r}_\parallel)\,\delta(\xi_n), \qquad \xi_n = (\boldsymbol{r}-\boldsymbol{r}_{S_n})\cdot\hat{\boldsymbol{n}}, \tag{31}$$

where $\boldsymbol{r}_\parallel$ is the tangential projection of $\boldsymbol{r}$ onto the plane of S_n and $\boldsymbol{r}_{S_n}$ is any reference point on S_n. The three-dimensional moments of $\tilde{\sigma}_n$ are then

$$\mu_n^{\boldsymbol{\alpha},S} \;\equiv\; \int_{\mathbb{R}^3} \tilde{\sigma}_n(\boldsymbol{r})\,(\boldsymbol{r}-\boldsymbol{c}_n)^{\boldsymbol{\alpha}}\,\mathrm{d}^3r \;=\; \int_{S_n} \sigma_n(\boldsymbol{r}_s)\,(\boldsymbol{r}_s-\boldsymbol{c}_n)^{\boldsymbol{\alpha}}\,\mathrm{d}^2r_s, \tag{32}$$

where the right-hand side is evaluated in the global Cartesian basis so that components along $\hat{\boldsymbol{n}}$ with $\alpha_\perp \geq 1$ automatically vanish and enter the linear system (34) below as the "thin-sheet constraint"

$$\sum_{q\in\mathcal{S}_n^S} \Lambda_{nq}^S\,\big(\boldsymbol{g}_q-\boldsymbol{c}_n\big)^{\boldsymbol{\alpha}} = 0 \quad \text{for all } \boldsymbol{\alpha} \text{ with } \alpha_\perp \geq 1, \tag{33}$$

forcing the projected point sources to collapse to a zero-thickness sheet in the grid's far-field representation.

Moment matching

In direct analogy with the bulk matching equation (28), the surface projection weights $\{\Lambda_{nq}^S\}_{q\in\mathcal{S}_n^S}$ are determined by

$$\sum_{q\in\mathcal{S}_n^S} \Lambda_{nq}^S\,(\boldsymbol{g}_q-\boldsymbol{c}_n)^{\boldsymbol{\alpha}} = \mu_n^{\boldsymbol{\alpha},S}, \qquad \forall\,|\boldsymbol{\alpha}| \leq n_{\mathrm{st}}, \tag{34}$$

where $\mathcal{S}_n^S$ is a cubic stencil of $M_{\mathrm{st}} = (n_{\mathrm{st}}+1)^3$ grid nodes centred on the triangle centroid $\boldsymbol{c}_n$. The stencil size and grid are identical to the bulk case of Sec. 5.1, so the surface and bulk contributions share a single FFT plan. Only a single scalar linear system per boundary face must be solved; the Vandermonde-type matrix $\big[(\boldsymbol{g}_q-\boldsymbol{c}_n)^{\boldsymbol{\alpha}}\big]$ is of size $\binom{n_{\mathrm{st}}+3}{3} \times M_{\mathrm{st}}$ and is independent of the kernel G, so it is factored once and reused for every boundary face.

Far-field product via the shared FFT

With the bulk vector weights $W_{np}^{\mathrm{vec},\alpha}$ (three Cartesian components), the bulk divergence weights $W_{np}^{\boldsymbol{\nabla}\cdot}$, and the new surface weights Λ_{nq}^S, the full far-field MVP $\boldsymbol{y}^{\mathrm{far}} = K^{\mathrm{far}}\boldsymbol{x}$ proceeds through a single grid pass. Define three source projections

$$q_p^{\mathrm{vec},\alpha} = \sum_n W_{np}^{\mathrm{vec},\alpha}\,x_n, \qquad q_p^{\boldsymbol{\nabla}\cdot} = \sum_n W_{np}^{\boldsymbol{\nabla}\cdot}\,x_n, \qquad q_p^S = \sum_{n\in\mathcal{N}_{\mathrm{bnd}}} \Lambda_{np}^S\,x_n, \tag{35}$$

where $\mathcal{N}_{\mathrm{bnd}}$ is the index set of half-SWG basis functions with a boundary face. A single three-dimensional FFT convolution with the Green-function kernel is then applied independently to each scalar source field,

$$Q_p^{\bullet} = \sum_{p'} G(\boldsymbol{g}_p - \boldsymbol{g}_{p'})\, q_{p'}^{\bullet}, \qquad \bullet \in \{\mathrm{vec}, \alpha;\ \boldsymbol{\nabla}\boldsymbol{\cdot};\ S\}, \tag{36}$$

reusing the precomputed FFT of G exactly as in Sec. 5.3. Finally, the test-side interpolation combines the vector and two scalar grid channels,

$$\begin{aligned} y_m^{\mathrm{far}} = k_0^2 \sum_{\alpha} \sum_{p\in\mathcal{S}_m} W_{mp}^{\mathrm{vec},\alpha}\, Q_p^{\mathrm{vec},\alpha} \;+\; \sum_{p\in\mathcal{S}_m} W_{mp}^{\boldsymbol{\nabla}\boldsymbol{\cdot}} \left(Q_p^S - Q_p^{\boldsymbol{\nabla}\boldsymbol{\cdot}}\right) \\ +\; \left[m \in \mathcal{N}_{\mathrm{bnd}}\right] \sum_{p\in\mathcal{S}_m^S} \Lambda_{mp}^S \left(Q_p^{\boldsymbol{\nabla}\boldsymbol{\cdot}} - Q_p^S\right), \end{aligned} \tag{37}$$

where the first line accumulates $K^{A,\mathrm{far}}$ and (via $Q^S - Q^{\boldsymbol{\nabla}\boldsymbol{\cdot}}$) $K^{B,\mathrm{far}}$ minus the divergence-divergence piece of K^A, and the second line adds $K^{C,\mathrm{far}}$ and $K^{D,\mathrm{far}}$ for test functions on a boundary face (Iverson bracket $[\,\cdot\,]$). The symmetric appearance of $Q^{\boldsymbol{\nabla}\boldsymbol{\cdot}} - Q^S$ on both sides reflects the charge-neutrality identity $Q_{\mathrm{vol}} + Q_{\mathrm{surf}} = 0$ of Sec. 3.3: the bulk divergence and the boundary-face normal flux are the two halves of a single conserved charge distribution, and the AIM projection preserves this cancellation on the grid.

No additional FFT is required compared to the bulk-only pipeline of Sec. 5.3: the surface channel is absorbed as one extra scalar grid array, for a total of $3 + 2 = 5$ scalar FFTs per MVP (three vector components, one bulk divergence, and one surface density) versus $3 + 1 = 4$ in Phase-1a.

Near-field correction

For pairs (m, n) with at least one of $\{m, n\}$ on a boundary face and $\|\boldsymbol{c}_m - \boldsymbol{c}_n\| \le r_{\mathrm{near}}$, the far-field approximation in Eq. (37) is insufficient. In direct analogy with the bulk precorrection of Sec. 5.4, we store the difference between the Duffy-integrated $K_{mn}^{B/C/D}$ of Sec. 3.2 and its AIM-reconstructed far-field value in a sparse correction block $K^{S,\mathrm{pc}}$. The storage scales as $\mathcal{O}(N_{\mathrm{bnd}}\,\langle M_{\mathrm{near}}^S\rangle)$, where $\langle M_{\mathrm{near}}^S\rangle$ is the mean number of near-neighbours per boundary face. For a mesh with characteristic edge length ℓ_c and near radius $r_{\mathrm{near}} = 2 \times$ (stencil extent), this count is $\mathcal{O}(1)$, so $K^{S,\mathrm{pc}} = \mathcal{O}(N_{\mathrm{bnd}})$. On compact objects with $N_{\mathrm{bnd}} = \mathcal{O}(N_{\mathrm{dof}}^{2/3})$, this is a reduction from $\mathcal{O}(N_{\mathrm{dof}}^{5/3})$ (`half_swg_extra` in Phase-1a) to $\mathcal{O}(N_{\mathrm{dof}}^{2/3})$—an asymptotic factor of N_{dof}.

Closed-form surface moment evaluation

The surface moment integral (32) over a planar triangle S_n admits a closed-form recursive evaluation in simplex coordinates, eliminating quadrature error entirely. Let $(\boldsymbol{r}_1, \boldsymbol{r}_2, \boldsymbol{r}_3)$ be the triangle vertices, (ξ, η) the affine simplex coordinates of the unit isosceles triangle $T_0 = \{0 \le \xi, \eta;\ \xi + \eta \le 1\}$, so that $\boldsymbol{r}_s = \boldsymbol{r}_1 + \xi(\boldsymbol{r}_2 - \boldsymbol{r}_1) + \eta(\boldsymbol{r}_3 - \boldsymbol{r}_1)$. Each monomial factor $(r_\beta - c_{n,\beta})$, $\beta \in \{x, y, z\}$, becomes linear in (ξ, η), $r_\beta - c_{n,\beta} = \mu_{10}^{(\beta)}\xi + \mu_{01}^{(\beta)}\eta + \mu_{00}^{(\beta)}$, and the order-$Q = |\boldsymbol{\alpha}|$ moment reduces to a bivariate polynomial integral

$$J_{\boldsymbol{\alpha}}^{(Q)} = \int_{T_0} \prod_{q=1}^{Q} \left(\mu_{10}^{(\beta_q)}\,\xi + \mu_{01}^{(\beta_q)}\,\eta + \mu_{00}^{(\beta_q)}\right) \mathrm{d}\xi\,\mathrm{d}\eta = \sum_{i=0}^{Q}\sum_{j=0}^{Q} \lambda_{ij}^{(Q)}\, \frac{i!\,j!}{(i+j+2)!}, \tag{38}$$

where β_q runs over the multi-index $\boldsymbol{\alpha}$ entry by entry (so Q factors appear in total). The polynomial coefficients $\lambda_{ij}^{(Q)}$ are generated by the recursion

$$\lambda_{ij}^{(Q+1)} = \lambda_{i-1,j}^{(Q)}\,\mu_{10}^{(\beta_{Q+1})} + \lambda_{i,j-1}^{(Q)}\,\mu_{01}^{(\beta_{Q+1})} + \lambda_{i,j}^{(Q)}\,\mu_{00}^{(\beta_{Q+1})}, \qquad \lambda_{00}^{(0)} = 1, \tag{39}$$

with $\lambda_{ij}^{(Q)} = 0$ whenever $i < 0$ or $j < 0$. The final surface moment is recovered by multiplying $J_{\boldsymbol{\alpha}}^{(Q)}$ by the Jacobian $2|S_n|$. The cost per boundary face is $\mathcal{O}(n_{\mathrm{st}}^4)$ operations with no numerical quadrature, eliminating one of the two truncation-error sources in the Phase-1a projection (the bulk tetrahedral moments retain a Gauss-cubature error of $\sim 10^{-12}$ which dominates machine epsilon but is well below the AIM far-field truncation at $n_{\mathrm{st}} = 2$).

Summary of Phase A impedance

With the surface projection in place, the complete AIM-accelerated impedance equation (29) is augmented to

$$\boldsymbol{y} = \left(D_M - \tfrac{\kappa}{\epsilon_{\mathrm{bg}}}\left(K^{\mathrm{far}} + K^{\mathrm{pc}} + K^{S,\mathrm{pc}}\right)\right)\boldsymbol{x}, \tag{40}$$

where K^{far} now evaluates all four kernel contributions $K^A + K^B + K^C + K^D$ via Eq. (37) and $K^{S,\mathrm{pc}}$ is the sparse surface-near-field correction. The Phase-1a `half_swg_extra` matrix is removed entirely.

Measured memory and timing

Table 1 reports the measured in-memory footprint of each `AIMOperator` component and the setup/solve wall times for the Au doublet benchmark ($R_{\mathrm{monomer}} = 30$ nm, gap 3 nm, $\lambda_0 = 638$ nm, Johnson–Christy [15] gold, Intel i7-1265U, 16 GB RAM, Julia 1.11, single-threaded pre-parallelization baseline — with the v0.5.x threaded setup and block MVP (Sec. 6.3) the `t_setup` and `t_solve` columns drop by a factor of 2–3 at 4+ Julia threads; script `benchmarks/cas_v2/doublet_mstm/pha` Each row corresponds to a refinement level $\ell_c = R/k$. For comparison, Phase-1a stored `half_swg_extra` as a dense $N_{\mathrm{bnd}} \times N_{\mathrm{dof}}$ sparse matrix which on the same geometry reached 486 MiB at $R/5$ and 1074 MiB at $R/6$; Phase A reduces those totals to 97 MiB and 164 MiB — a 5.0× / 6.5× / 11× reduction at $R/5$ / $R/6$ / $R/7$ respectively.

Table 1: Phase A Au doublet memory (in-memory footprint via `Base.summarysize`) and wall times (3-orientation block-BiCGSTAB solve). Measured on commit `81c7961`.

ℓ_c	N_{dof}	N_{bnd}	W^S [MiB]	K^{pc} [MiB]	total [MiB]	t_{setup} [s]	t_{solve} [s]
$R/5$	11 501	1610	4.9	67.2	96.8	105	201
$R/6$	18 919	2254	8.1	116.5	164.2	179	363
$R/7$	29 636	3176	12.6	189.0	262.7	299	718
$R/8$	45 585	4210	19.4	300.7	413.3	410	925

The total in-memory footprint grows as $\mathcal{O}(N_{\mathrm{dof}}^{1.03})$ between $R/5$ and $R/7$ — essentially linear in N_{dof}, confirming the Phase A design target. The dominant contribution is the bulk precorrection block K^{pc} (69 % to 72 % of total), whose sparsity scales as $\mathcal{O}(N_{\mathrm{dof}}\, n_{\mathrm{near}})$ with $n_{\mathrm{near}} \lesssim (M_{\mathrm{st}}\, r_{\mathrm{near}}/\ell_c)^3$ an $\mathcal{O}(1)$ count independent of N_{dof}. The new Phase A surface projection W^S adds 4 % to 5 % to the total, a small overhead compared with the memory eliminated.

The setup and solve times both scale as $\mathcal{O}(N_{\mathrm{dof}})$ within measurement noise, consistent with the AIM MVP cost $\mathcal{O}(N_g \log N_g + N_{\mathrm{dof}}\, n_{\mathrm{near}})$ evaluated at fixed wavelength. With sub-300 MiB budgets through $R/7$, further refinement to $R/8$ and $R/10$ is feasible on a 16 GB workstation and is the next step toward closing the residual 5.6 % Im $S_{\mathrm{fw},\theta}$ error at $\beta = \pi/2$ (Sec. 8.5).

6 Solvers

6.1 Direct and Krylov solvers

For small problems ($N_{\rm dof} \lesssim$ a few thousand), the dense assembly of Z and its LU factorisation is practical and serves as a validation reference for the AIM path. For larger problems we use a Krylov iterative method (BiCGSTAB from `Krylov.jl` [12]) applied to the AIM operator (29). A Jacobi preconditioner formed from the diagonal of the sparse mass and precorrection contributions suffices in our tests.

6.2 Multi-orientation batch solve

Because the impedance matrix Z depends only on the particle geometry and permittivity, it can be assembled once and then reused across many orientations: each orientation changes only the right-hand side $\boldsymbol{b}$. The function `solve_cas_v2_orientations` is the user-facing entry point; by default it selects `method=:aim_bicgstab`, which builds the AIM operator (Sec. 5) once and then solves the block system $\mathbf{ZX} = \mathbf{B}$ for all orientations simultaneously via a single block-Krylov iteration (Sec. 10). For small problems ($N_{\rm dof} \lesssim 10^3$) `method=:dense` falls back to the textbook assemble-Z-and-LU-factorise path, back-substituting for every $(\alpha_\ell, \beta_\ell, \gamma_\ell)$ in the input list. For spheroids either route is especially efficient: only the $\alpha = 0$ orientation needs an independent solve for each β, and the remaining α values are obtained by the analytical expansion of Sec. 8.3.

6.3 Parallel execution

Every computationally significant loop in the setup, MVP, and post-processing paths is parallelised for shared-memory multi-core execution. The user controls the number of worker threads with `julia -t N` (or the `JULIA_NUM_THREADS` environment variable); no other configuration is required.

Setup. The precorrection assembly (Sec. 5.4) is the dominant setup cost for any realistic problem — $\gtrsim$90 % of `build_aim_operator` wall time at $N_{\rm dof} \gtrsim 10^3$. Its outer near-column loop is partitioned into chunks and launched as independent `Threads.@spawn` tasks; each task accumulates its own COO triplets and the resulting `SparseMatrixCSC` is built once from the concatenated lists. The same pattern parallelises `build_aim_projection` (basis loop) and the near-field impedance assembly used by the dense path. BLAS is forced to a single thread inside the parallel region to avoid nested oversubscription from LAPACK calls (the monomial-matrix least-squares solves inside `build_aim_projection`). On an Intel Xeon 20-core machine against a $N_{\rm swg} = 3041$ sphere mesh, `assemble_precorrection` drops from 25.9 s (1 thread) to 10.5 s (4 threads, 2.5×) and 8.9 s (10 threads, 2.9×).

Block MVP. The column loop in `aim_mvp(op, X)` is parallelised: each of the L right-hand-side columns is dispatched as its own task, so for a machine with $N_{\rm thr}$ Julia threads the FFT work of the block MVP runs at effective width $\min(L, N_{\rm thr})$. Details in Sec. 10.

FFT. FFTW maintains its own thread pool; at module load `BlockVIEM` sets it to `Threads.nthreads()` via `FFTW.set_num_threads`. The exported `BlockVIEM.set_fft_threads(n)` resets it at run time. The recommended balance for a block MVP with L columns is

$$n_{\rm FFT} \;=\; \max\bigl(1,\; \lceil N_{\rm thr}/L \rceil\bigr), \tag{41}$$

so that the column and FFT parallelism together saturate the cores without oversubscription.

Parameter-sweep reuse. The projection matrices $\mathbf{W}_{x,y,z}, \mathbf{W}_{\mathrm{div}}, \mathbf{W}_{\mathrm{surf}}$ and the mass matrix $\mathbf{M}$ depend only on the mesh, the grid pitch and padding, and the polynomial order — not on k_0 or ϵ_p. The `build_aim_operator` constructor accepts keyword arguments `projection` and `mass` so that wavelength or material sweeps can build those pieces once and reuse them, rebuilding only the Green-function FFT kernel and the precorrection for each new (k_0, ϵ_p). The MVP is bit-identical between the fresh and reused paths (tolerance $< 10^{-13}$).

7 Scattering Observables

Let D_n $(n = 1, \ldots, N_{\mathrm{dof}})$ denote the solution of (12) and $\boldsymbol{D}(\boldsymbol{r}) = \sum D_n\, \boldsymbol{f}_n(\boldsymbol{r})$ the reconstructed electric flux density. This section collects the observables computed from D_n that correspond to the CAS-v2 detector protocol [11] and can be compared directly with `block-DDA_Py`.

7.1 Far-field scattering amplitude

The scattered far field in the physics convention reads

$$\boldsymbol{E}^{\mathrm{sca}}(\boldsymbol{r}) \sim \frac{\mathrm{e}^{+ik_0 r}}{4\pi\, r}\, \boldsymbol{F}(\hat{\boldsymbol{r}}), \qquad r \to \infty, \tag{42}$$

where the vector scattering amplitude is obtained by transverse projection of the Fourier transform of $\boldsymbol{D}$:

$$\boldsymbol{F}(\hat{\boldsymbol{r}}) = \frac{k_0^2\, \kappa}{\epsilon_{\mathrm{bg}}} \big(\mathbf{I}_3 - \hat{\boldsymbol{r}} \otimes \hat{\boldsymbol{r}}\big) \cdot \sum_{n=1}^{N_{\mathrm{dof}}} D_n \int_{\Omega} \boldsymbol{f}_n(\boldsymbol{r}')\, \mathrm{e}^{-ik_0\, \hat{\boldsymbol{r}} \cdot \boldsymbol{r}'}\, \mathrm{d}^3 r'. \tag{43}$$

The standard scattering amplitude used in Bohren–Huffman [10] is $\boldsymbol{f}(\hat{\boldsymbol{r}}) = \boldsymbol{F}(\hat{\boldsymbol{r}})/(4\pi)$.

7.2 Absorption cross section

The absorption cross section is evaluated directly from the volumetric Joule-loss formula, which is bilinear in $\boldsymbol{D}$ and hence reduces to a quadratic form in the expansion coefficients:

$$C_{\mathrm{abs}} = \frac{k_0\ \mathrm{Im}(\epsilon_p)}{\epsilon_{\mathrm{bg}}\, |\epsilon_p|^2\, |\boldsymbol{E}_0|^2}\ \mathrm{Re}\big(\boldsymbol{D}^{\dagger} \mathbf{M}\, \boldsymbol{D}\big), \tag{44}$$

where $\mathbf{M}$ is the mass matrix (23). For real ϵ_p this identically vanishes.

7.3 Scattering cross section by far-field integration

The scattering cross section is computed as the integral of the radiated flux density over the unit sphere,

$$C_{\mathrm{sca}} = \frac{1}{16\pi^2\, |\boldsymbol{E}_0|^2} \int_{S^2} |\boldsymbol{F}(\hat{\boldsymbol{r}})|^2\, \mathrm{d}\Omega, \tag{45}$$

using a product Gauss–Legendre (in $\cos\theta$) times trapezoid (in ϕ) rule with $2n_\theta^2$ directions. The default $n_\theta = 10$ is sufficient for size parameters $k_0\, r_{ve} \lesssim 5$; larger targets use $n_\theta \geq 20$. The quadratic functional $|\boldsymbol{F}|^2$ is nonnegative and insensitive to the sign-convention subtleties that plague forward-direction optical-theorem estimates of C_{ext}, so in practice we obtain C_{ext} from

$$C_{\mathrm{ext}} = C_{\mathrm{abs}} + C_{\mathrm{sca}}. \tag{46}$$

7.4 PCAS forward amplitude

The CAS-v2 forward scattering observables are defined with a circular incident polarisation [11]. Let $\hat{\boldsymbol{u}}_{\text{fw}}$ be the forward scattering direction (equal to the incident propagation direction) and $\hat{\boldsymbol{e}}_{\theta,\text{fw}}, \hat{\boldsymbol{e}}_{\phi,\text{fw}}$ the associated polar and azimuthal unit vectors in the particle frame. The s- and p-channel PCAS forward observables are

$$S_{\text{fw},\theta}^{\text{PCAS}} \; [= S_s(0)] = \frac{\sqrt{2}}{4\pi} \boldsymbol{F}(\hat{\boldsymbol{u}}_{\text{fw}}) \cdot \hat{\boldsymbol{e}}_{\theta,\text{fw}}, \tag{47}$$

$$S_{\text{fw},\phi}^{\text{PCAS}} \; [= S_p(0)] = -\frac{i\sqrt{2}}{4\pi} \boldsymbol{F}(\hat{\boldsymbol{u}}_{\text{fw}}) \cdot \hat{\boldsymbol{e}}_{\phi,\text{fw}}, \tag{48}$$

where the $\sqrt{2}$ and $i\sqrt{2}$ factors are inherited from the circular polarisation decomposition (6), and the $1/(4\pi)$ factor converts from $\boldsymbol{F}$ to the BH83-normalised scattering amplitude $\boldsymbol{f} = \boldsymbol{F}/(4\pi)$. Equations (47) and (48) are the two physical quantities measured by the PCAS instrument. For convenience, their arithmetic mean

$$S_{\text{fw,mean}}^{\text{PCAS}} = \frac{S_{\text{fw},\theta}^{\text{PCAS}} + S_{\text{fw},\phi}^{\text{PCAS}}}{2} \tag{49}$$

is also stored in the `CASv2Result` container as a derived single-channel summary; (47) and (48) remain the primary observables.

7.5 OCBS backward amplitude

For the exact backward scattering direction $\hat{\boldsymbol{u}}_{\text{bk}} = -\hat{\boldsymbol{u}}_{\text{inc}}$, the laboratory basis is $\hat{\boldsymbol{e}}_{\theta,\text{bk}} = -\hat{\boldsymbol{e}}_{\theta,\text{inc}}$ and $\hat{\boldsymbol{e}}_{\phi,\text{bk}} = +\hat{\boldsymbol{e}}_{\phi,\text{inc}}$. The θ- and ϕ-channel backward amplitudes are defined in direct analogy with (47)–(48) at $\hat{\boldsymbol{r}} = \hat{\boldsymbol{u}}_{\text{bk}}$, and the single observable reported by the OCBS (Optical Coherence Backscattering Sensor) instrument [11] is the linear combination

$$S_{\text{bk}}^{\text{OCBS}} = \frac{-S_{\text{bk},\theta}^{\text{OCBS}} + S_{\text{bk},\phi}^{\text{OCBS}}}{\sqrt{2}}. \tag{50}$$

As shown in [11], combining the two channels with this sign pattern cancels the off-diagonal Mishchenko elements $S_{12}(\pi) + S_{21}(\pi) = 0$ and produces a single, orientation-stable scalar observable.

8 Validation

8.1 Mie sphere

For a homogeneous sphere the exact analytical solution is the Mie series [10]. The reference implementation in `test/mie_reference.jl` provides C_{ext}, C_{sca}, C_{abs}, and the full complex amplitude matrix elements S_{11}, S_{22} at $\theta = 0, \pi$. The Phase-5 validation tests compare the VIEM solution to this Mie reference on a volume-equivalent sphere with $\lambda_0 = 4\,\mu\text{m}$, $m_m = 1$, $m_p = 1.5 + 0.01i$, $r = 0.5\,\mu\text{m}$ and mesh sizes $\ell_c \in \{0.30, 0.18, 0.12\}$. After the 2026-04-13 physics-convention switch (Sec. 7.1), both real and imaginary parts of the PCAS forward amplitude (49) agree with the Mie reference to $\sim 0.4\%$ at the coarsest mesh and $\sim 0.1\%$ at $\ell_c = 0.12$.

8.2 Plasmonic gold sphere (highly absorbing)

The dielectric tests above use a weakly absorbing dielectric ($\operatorname{Im} m_p \ll \operatorname{Re} m_p$). To stress the half-SWG formulation in a qualitatively different regime, we run a Mie comparison on a plasmonic gold sphere with the Johnson & Christy [15] optical constants (via `https://refractiveindex.info`)

$$m_{\mathrm{Au}} = 0.17525 + 3.4830\,i \quad (\lambda_0 = 0.638\,\mu\mathrm{m}), \qquad \varepsilon_{\mathrm{Au}} = -12.10 + 1.22\,i. \tag{51}$$

The negative real part of $\varepsilon_{\mathrm{Au}}$ places the problem in the plasmonic regime where the bulk polarisation current and the surface polarisation charge are both large and must mostly cancel each other in the divergence; capturing both contributions correctly is the test of the half-SWG surface terms $K^B + K^C + K^D$ derived in Sec. 3. The radius is fixed at the size parameter $x = k_0 r = 0.63$, giving $r \approx 0.0640\,\mu\mathrm{m}$, a typical AuNP scale. Mie reference values for the target sphere are

$$Q_{\mathrm{ext}} = 1.386, \quad Q_{\mathrm{abs}} = 0.192, \quad Q_{\mathrm{sca}} = 1.195,$$
$$S^{\mathrm{PCAS}}_{\mathrm{fw,mean}} = +0.03877 + 0.01397\,i, \qquad S^{\mathrm{OCBS}}_{\mathrm{bk}} = +0.05946 + 0.01907\,i.$$

The VIEM solution is computed with circular incident polarisation $\hat{\boldsymbol{e}}_0 = (\hat{\boldsymbol{x}} + i\hat{\boldsymbol{y}})/\sqrt{2}$ at two mesh refinements; results are compared to Mie evaluated at the volume-equivalent radius of each mesh (so that the residual mesh volume error does not contaminate the discretisation error). Cross sections are obtained via `compute_scattering` (absorption from the volumetric Joule integral, scattering by optical theorem) and the polarimetric amplitudes via `compute_cas_observables`.

Table 2: Mie validation of `block-VIEM.jl` on a plasmonic gold sphere. $m_{\mathrm{Au}} = 0.17525 + 3.4830\,i$ (Johnson & Christy 1972, via `https://refractiveindex.info`), $\lambda_0 = 0.638\,\mu\mathrm{m}$, $x = k_0 r = 0.63$, $r \approx 0.0640\,\mu\mathrm{m}$. Errors are relative to Mie evaluated at the actual mesh volume-equivalent radius; the cross-section column shows $|\Delta C|/C_{\mathrm{Mie}}$, the amplitude columns $|\Delta S|/|S|_{\mathrm{Mie}}$. Both dense (LU) and AIM-BiCGSTAB solvers are shown. Generated by `benchmarks/cas_v2/au_sphere_mie.jl` with half-SWG basis functions and `duffy_reference_rule(5)`.

ℓ_c [µm]	N	Method	C_{ext} [%]	C_{abs} [%]	C_{sca} [%]	$S_{\mathrm{fw,mean}}$ [%]	S_{bk} [%]
0.020	2134	dense (LU)	0.65	0.005	0.75	0.58	0.28
0.020	2134	AIM-BiCGSTAB	0.25	0.44	0.37	0.32	0.11
0.014	4932	dense (LU)	0.34	0.08	0.41	0.32	0.15
0.014	4932	AIM-BiCGSTAB	0.15	0.24	0.22	0.18	0.07

Several observations follow from Table 2.

1. All five observables converge *monotonically* between the two mesh resolutions, and the per-DoF improvement is roughly consistent with h^2 scaling — doubling the DoF count from 2134 to 4932 (a factor $\sim 2.3\times$) cuts every error by a factor ~ 2.

2. Even at the coarse mesh ($N = 2134$, only ~ 4 elements per radius) every observable is below 1 % relative error, which is well inside the target accuracy for CAS-v2 LUT generation (the downstream Bayesian retrieval typically tolerates $\sim 2\%$ on the polarimetric amplitudes).

3. The polarimetric amplitudes $S^{\mathrm{PCAS}}_{\mathrm{fw,mean}}$ and $S^{\mathrm{OCBS}}_{\mathrm{bk}}$ are *more accurate* than the optical-theorem cross sections at the same mesh, because the cross-section evaluation involves an additional far-field integral on the unit sphere whose quadrature error adds to the SWG discretisation error. The amplitudes use only a single evaluation of $\boldsymbol{F}$ at the forward / backward direction and inherit only the SWG error.

4. The fact that errors are monotone and small in the plasmonic regime — where the divergence-free part of the current density is comparable in magnitude to the divergent (charge) part — confirms that the half-SWG surface correction $K^B + K^C + K^D$ correctly enforces the surface polarisation-charge boundary condition. The interior-only SWG basis (without the boundary half-functions) would produce 10–30 % errors here, identical to the failure mode documented in Sec. 3.3 for weakly-absorbing dielectrics.

Combined validation summary. Across the dielectric ($m_p = 1.5 + 0.01\,i$, Sec. 8.1) and plasmonic ($m_p = 0.17525 + 3.4830\,i$, this section) Mie tests, the block-VIEM.jl half-SWG solver achieves $\leq 1\,\%$ relative error on C_{ext}, C_{abs}, C_{sca}, $S^{\text{PCAS}}_{\text{fw,mean}}$ and $S^{\text{OCBS}}_{\text{bk}}$ already at modest mesh sizes ($N \sim 2 \times 10^3$ DoF), and the errors decay monotonically with refinement.

8.3 Oblate spheroid α-symmetry

A prolate or oblate spheroid has a continuous C_∞ rotation symmetry about its figure axis. When the orientation is parameterised by intrinsic ZYZ Euler angles with $\gamma = 0$, rotating the first angle α does not change the physical orientation of an axisymmetric particle in the particle frame; it only rotates the incident polarisation basis by α about $\hat{\boldsymbol{u}}_{\text{inc}}$. For the CAS-v2 observables this implies the analytical α-expansion [11]

$$S_\theta(\alpha, \beta, \gamma{=}0) = A(\beta) + B(\beta)\,\mathrm{e}^{+2i\alpha}, \tag{52}$$

$$S_\phi(\alpha, \beta, \gamma{=}0) = A(\beta) - B(\beta)\,\mathrm{e}^{+2i\alpha}, \tag{53}$$

with

$$A(\beta) = \tfrac{1}{2}\big[S_\theta(0, \beta, 0) + S_\phi(0, \beta, 0)\big], \quad B(\beta) = \tfrac{1}{2}\big[S_\theta(0, \beta, 0) - S_\phi(0, \beta, 0)\big]. \tag{54}$$

The identity (52)–(53) is the exact basis of the α-analytical expansion that `block-DDA_Py` uses to reduce the orientation sweep to solves at $\alpha = 0$ only. Because VIEM after the physics-convention switch lives in the same time-convention as block-DDA_Py, the sign of the exponent is $+2i\alpha$ — identical to block-DDA_Py.

The benchmark script `benchmarks/cas_v2/spheroid_ar3.jl` builds an oblate spheroid with bc_ratio $= b/c = 3$, solves at $\beta = \pi/4$ for $\alpha \in \{0, \pi/8, \pi/4, 3\pi/8, \pi/2\}$, and compares the independently solved amplitudes to the analytical prediction (52)–(53). The maximum relative deviation is $\sim 5 \times 10^{-15}$, that is, machine precision for double-precision arithmetic.

The α-symmetry check verifies the internal consistency of block-VIEM.jl but does not confirm that the computed amplitudes themselves are correct. A direct cross-validation against block-DDA_Py on the same oblate spheroid is therefore performed by the driver pair under `benchmarks/cas_v2/dda_comparison/` (Python for DDA, Julia for VIEM, JSON as the exchange format):

- particle: oblate spheroid, $D_{ve} = 0.40\,\mu\text{m}$, bc_ratio = 3, $m_p = 1.5$ (non-absorbing);
- background: $m_m = 1$, $\lambda_0 = 0.638\,\mu\text{m}$ ($k_0 r_{ve} \approx 1.97$);
- orientations: two tilt angles, $(\alpha, \beta, \gamma) = (0, \pi/4, 0)$ and $(0, \pi/2, 0)$;
- discretisations: DDA uses $N_{\text{dip}} = 2236$ dipoles with dpl = 17; VIEM uses a half-SWG mesh with $\ell_c = 0.035\,\mu\text{m}$ yielding $N_{\text{dof}} = 8746$.

The PCAS forward amplitudes agree to within $\sim 3\%$ at both tilt angles:

The two tilt angles cover qualitatively different illumination geometries: at $\beta = \pi/4$ the spheroid is intermediate between flat-on and edge-on; at $\beta = \pi/2$ the figure axis is exactly perpendicular to $\hat{\boldsymbol{u}}_{\text{inc}}$ (edge-on), so the incident wave sees the largest geometrical cross section and the C_{ext} value is roughly 9% larger than at $\pi/4$. The fact that VIEM and DDA agree to the same few-percent level at both tilts — not just on the magnitude but on both the real and imaginary parts of $S_{\text{fw},\theta}$ and $S_{\text{fw},\phi}$ independently — demonstrates that the agreement is not an accidental near-coincidence at one orientation but a genuine convergence to the same physical scattering amplitudes. Both codes are discretisation-limited at these resolutions — the VIEM volume-equivalent radius recovered from the mesh is $r_{ve}^{\text{mesh}} = 0.199\,18\,\mu\text{m}$, already a 0.41% volumetric error on top of the half-SWG error budget, and DDA has its own stair-casing error of comparable size — so a $\sim 2\%$ disagreement between the two independent codes is the level

Table 3: Cross-validation of VIEM against block-DDA_Py on a single oblate AR = 3 spheroid at two tilt angles. The relative error is $|\Delta S|/|S_{\mathrm{DDA}}|$.

Observable	block-DDA_Py	block-VIEM.jl	rel. err.
$\beta = \pi/4$			
$S_{\mathrm{fw},\theta}$	$+0.23514 + 0.09368i$	$+0.23143 + 0.09164i$	1.67%
$S_{\mathrm{fw},\phi}$	$+0.23884 + 0.18561i$	$+0.23480 + 0.17937i$	2.46%
C_{ext}	$0.178\,19\,\mu\mathrm{m}^2$	$0.172\,91\,\mu\mathrm{m}^2$	2.96%
$\beta = \pi/2$			
$S_{\mathrm{fw},\theta}$	$+0.23703 + 0.08742i$	$+0.23221 + 0.08536i$	2.08%
$S_{\mathrm{fw},\phi}$	$+0.29578 + 0.21695i$	$+0.28980 + 0.20883i$	2.75%
C_{ext}	$0.194\,19\,\mu\mathrm{m}^2$	$0.187\,69\,\mu\mathrm{m}^2$	3.34%

of agreement one can expect at these mesh densities. The result confirms that block-VIEM.jl and block-DDA_Py produce the *same complex polarimetric amplitudes* on an axisymmetric non-spherical particle up to discretisation error, in addition to reproducing the Mie sphere limit to $\sim 0.4\%$ (Sec. 8.1).

8.4 Accuracy and computational cost vs. block-DDA_Py

Table 4 compares the C_{abs} error produced by block-VIEM.jl (half-SWG on a tetrahedral mesh) and block-DDA_Py (cubic-lattice DDA with the Chaumet–Rahmani dynamic polarisability) on the same homogeneous Mie sphere, $m_p = 1.5 + 0.01i$, $r = 1\,\mu\mathrm{m}$, $\lambda_0 = 10\,\mu\mathrm{m}$ (size parameter $x \approx 0.63$).

Table 4: Per-DoF accuracy comparison on a Mie sphere ($m_p = 1.5 + 0.01i$, $x \approx 0.63$). The VIEM solves use the half-SWG basis and Duffy volumetric quadrature; the DDA solves use block-BiCGSTAB on the Goodman–Draine–Flatau FFT operator.

Method	N_{dof}	C_{abs} rel. error
DDA, spacing $d = 0.14\,\mu\mathrm{m}$	4488	1.25%
half-SWG VIEM, $\ell_c = 0.50\,\mu\mathrm{m}$	589	0.21%
half-SWG VIEM, $\ell_c = 0.18\,\mu\mathrm{m}$	7868	0.15%

For this test block-VIEM.jl is roughly **10–12$\times$ more accurate per DoF** than block-DDA_Py: matching the 1.25% DDA error requires only ~ 400 half-SWG DoFs (versus ~ 4500 DDA dipoles), and at the same DoF count (~ 8000) the VIEM error is an order of magnitude smaller.

There are four mutually reinforcing reasons.

1. *Conforming geometry.* A tetrahedral mesh represents curved boundaries to $O(\ell_c^2)$, whereas a cubic lattice stair-cases them to $O(d)$. For Mie spheres this alone removes a geometric contribution to C_{abs} and C_{sca} that scales as d rather than d^2.

2. *Conforming vector space.* The SWG/half-SWG basis is $H(\mathrm{div})$-conforming, so the normal component of $\boldsymbol{D}$ is automatically continuous across tetrahedron faces — the physically correct boundary condition at a dielectric interface. Expanding $\boldsymbol{D}$ instead of $\boldsymbol{E}$ [1, Sec. II.B] further ensures that the polarisation jump at $\partial\Omega$ is represented exactly by the half-SWG functions on the boundary faces, with the charge-neutrality identity of Sec. 3.3. In DDA the induced polarisation is piecewise constant at the dipole centres, so the interface condition is only recovered in the limit $d \to 0$.

3. *Linear vector basis vs point dipoles.* Within each tetrahedron $\boldsymbol{D}$ is represented by up to four SWG functions, each linear in position and with a piecewise-constant divergence. DDA represents the polarisation by a single point dipole per element, i.e. a delta distribution localised at the lattice node. The tetrahedral moment approximation is therefore intrinsically one order higher, and in the far field the error per element scales as $k_0 h$ rather than $(k_0 d)^0$.

4. *Physical polarisability.* DDA requires a dynamic polarisability prescription (Clausius–Mossotti plus the $(2/3)(ka)^3$ radiative correction [8]) to cancel leading-order interaction errors, and the choice of prescription feeds back into the error. VIEM solves the integral equation directly; $\boldsymbol{D}$ is the physical variable, and no such tuning is required.

Against these accuracy advantages, block-VIEM.jl pays a higher per-DoF setup cost. Table 5 summarises the asymptotic cost and memory for a matrix-free MVP:

Table 5: Asymptotic cost model. N is the number of unknowns, N_g the number of AIM or FFT grid points (typically $N_g \sim 2N$ for AIM), n_{st} the AIM moment matching order, n_{it} the number of Krylov iterations, and L the number of particle orientations in a batch solve.

Quantity	block-DDA_Py	block-VIEM.jl (half-SWG + AIM)
Element geometry	cubic lattice	unstructured tetrahedra
Green kernel on grid	analytic dyadic, $\mathcal{O}(N_g)$	scalar + moment-matched dyadic, $\mathcal{O}(N_g)$
Near-field "precorrection"	none (exact on the lattice)	Duffy volumetric, $\mathcal{O}(N\, n_{\mathrm{near}})$
MVP cost	$\mathcal{O}(N_g \log N_g)$	$\mathcal{O}(N_g \log N_g + N\, n_{\mathrm{near}})$
Multi-orientation batch	block-BiCGSTAB on L RHS	shared LU (small N) or L indep. BiCGSTAB
Dominant memory	$\mathcal{O}(N_g)$ FFT kernel	$\mathcal{O}(N_g)$ FFT kernel $+\, \mathcal{O}(N\, n_{\mathrm{near}})$ precorr.

The central difference is that DDA has a strictly translation-invariant interaction matrix, so a single FFT on the doubled lattice captures the entire MVP exactly. VIEM has no such property: each basis function sees a different discrete support of neighbouring tetrahedra, and the AIM moment matching is only accurate for pairs whose separation is large compared with the element size. Near pairs must therefore be computed directly via Duffy quadrature and stored as a sparse precorrection matrix (Sec. 5.4). For typical settings, this near-field work dominates both the matrix assembly time and the per-iteration MVP cost at small to moderate N; only for very large meshes does the $N_g \log N_g$ FFT cost become comparable.

The overall trade-off is therefore: **VIEM spends more work per unknown (near-field Duffy integrals and precorrection), but uses an order of magnitude fewer unknowns to reach a given accuracy for high-contrast particles with curved boundaries.** For sphere-like, low-contrast problems DDA's simplicity wins on wall-clock time; for the high-contrast iron-oxide aggregates and faceted gold nanoparticles targeted by this project, VIEM is the clear winner and the block-VIEM.jl $\ell_c = 0.50$ line of Table 4 is representative of production use.

8.5 Two-sphere doublet vs MSTM (multi-orientation)

A second independent cross-validation compares block-VIEM.jl against the Multi-Sphere T-Matrix package MSTMforCAS.jl [17], which produces numerically exact CAS-v2 observables for aggregates of homogeneous spheres by expanding the internal and scattered fields in vector spherical wave functions (VSWFs) and solving the multi-sphere interaction equation exactly.

This tests block-VIEM.jl on a genuinely multi-body target and, because the geometry is non-axisymmetric when the incidence is off-axis, exercises the rotational-symmetry machinery of the CAS-v2 observable.

Geometry. Two equal-radius spheres of radius $R = 0.030\,\mu\mathrm{m}$ with a small axial gap of $0.003\,\mu\mathrm{m}$ (centre-to-centre separation $d = 0.063\,\mu\mathrm{m}$) lie along the particle-frame z-axis. The spheres are topologically disjoint, allowing the MSTM code to treat each monomer as an isolated VSWF expansion. The vacuum wavelength is $\lambda_0 = 0.638\,\mu\mathrm{m}$ and the background medium is vacuum, $m_m = 1$.

Orientations. Let β denote the angle between the incidence direction and the doublet axis. In block-VIEM.jl the particle is kept fixed along the lab z-axis and the incidence direction is rotated via the Euler-angle triple $(\alpha, \beta, \gamma) = (0, \beta, 0)$ through the VIEM rotation machinery (Sec. 7.4). In MSTMforCAS.jl, which fixes the incidence along $+\hat{\boldsymbol{z}}$, the doublet axis is rotated about $\hat{\boldsymbol{e}}_y$ so that it makes the same angle β with the z-axis. The two setups are physically equivalent.

Observable. The CAS-v2 forward amplitudes are defined in the block-DDA_Py / block-VIEM convention (RCP incidence) as

$$S_{\mathrm{fw},\theta} = S_{11} + i\,S_{12}, \quad S_{\mathrm{fw},\varphi} = S_{22} - i\,S_{21}, \quad S_{\mathrm{fw,mean}} = \tfrac{1}{2}\big(S_{\mathrm{fw},\theta} + S_{\mathrm{fw},\varphi}\big), \tag{55}$$

where S_{ij} are the MI02 scattering amplitudes at the forward direction (Sec. 7.4). The block-VIEM.jl side returns $S_{\mathrm{fw,mean}}$ directly from `compute_cas_observables`. The MSTMforCAS.jl side returns the BH83 forward amplitude tuple (S_1, S_2, S_3, S_4), which is first converted to MI02 via

$$S_{11} = \frac{S_2}{-ik_m}, \quad S_{12} = \frac{S_3}{ik_m}, \quad S_{21} = \frac{S_4}{ik_m}, \quad S_{22} = \frac{S_1}{-ik_m}, \tag{56}$$

and then substituted into (55) to obtain

$$S_{\mathrm{fw},\theta} = \frac{S_2 - i\,S_3}{-ik_m}, \qquad\qquad S_{\mathrm{fw},\varphi} = \frac{S_1 + i\,S_4}{-ik_m}, \tag{57}$$

with $k_m = 2\pi m_m/\lambda_0$. The sign flip on the off-diagonal entries in (56) comes from the opposite handedness of the MI02 $\hat{\boldsymbol{e}}_\varphi$ basis vector relative to the BH83 $\hat{\boldsymbol{e}}_\perp$. For the x-z-plane-symmetric doublet used in this benchmark one has $S_3 = S_4 = 0$, so the off-diagonal terms drop out; the correct signs are retained in the driver code for generic (symmetry-free) aggregates.

Numerical setup. The block-VIEM.jl discretisation uses $\ell_c = R/5 = 0.006\,\mu\mathrm{m}$, giving $n_{\mathrm{tet}} = 5348$ tetrahedra and $N = 11501$ SWG degrees of freedom (half-SWG, boundary faces included). The AIM block-BiCGSTAB solver is run with tolerance 10^{-7} and pitch $0.5\bar{h} = 0.0038\,\mu\mathrm{m}$ (Sec. 10).

MSTMforCAS.jl's automatic VSWF truncation for $x_{\mathrm{monomer}} \approx 0.30$ returns $N = 3$, based on a single-sphere Mie convergence check (partial Q_{ext} within 10^{-6}). That choice is converged for the high-index dielectric but not for gold: the product $|m \cdot x| \approx 1.03$ combined with the inter-sphere translation coupling amplifies higher-order multipoles, and $|S_{\mathrm{fw,mean}}|$ shifts by $\approx 5\%$ between $N = 3$ and the fully-converged $N = 15$ reference (a geometric convergence with per-order ratio ≈ 0.3). We therefore force $N_{\mathrm{trunc}} = 15$ in `run_mstm.jl` for both materials: computationally cheap for a 2-sphere aggregate ($\lesssim 0.5\,\mathrm{s}$ per orientation) and delivers a reference with rtol $\leq 10^{-6}$.

MSTMforCAS $\geq$ 0.4.3 incorporates three numerical-stability fixes contributed as part of this benchmark work:

1. Miller's downward recurrence for the Riccati–Bessel $\psi_n(x)$ replaces the conditionally-stable upward recurrence (Wiscombe 1980); the upward recurrence previously failed catastrophically at $N \geq 7$ for $x \approx 0.3$ because the underflow $\psi_n(x) \sim x^n/(2n-1)!!$ reaches $\sim 10^{-9}$ at $n = 8$ and is drowned by the spurious χ_n-component amplification factor $(2n-1)/x \approx 44$ per step.

2. Correct sizing of the per-sphere Mie-coefficient buffer `mie_vecs[i]` to `nois[i]` when the user-requested `truncation_order` exceeds the default Wiscombe bound `mie_nmax(x)`.

3. Miller's downward recurrence for the spherical-Bessel $j_n(z)$ used in the multi-sphere translation operator, eliminating the analogous $y_n(kd) \sim (2n-1)!!\,(kd)^{-n-1}$ precision loss at small kd.

With these fixes MSTMforCAS converges geometrically in N to $\leq 10^{-6}$ by $N = 15$ for both materials at $x \approx 0.3$; the `check_trunc_convergence.jl` script verifies this at each benchmark point. CBICG tolerance 10^{-8}.

Results. The benchmark is run at three orientations $\beta \in \{0, \pi/4, \pi/2\}$ and two refractive indices: a high-contrast dielectric $m_p = 1.60 + 0.01i$ (polystyrene-like) and plasmonic gold $m_p = 0.17525 + 3.4830i$ at 638 nm (Johnson & Christy 1972). For the dielectric case the relative error of $|S_{\text{fw,mean}}|$ stays within 1.6 % across all orientations. For the plasmonic gold doublet the error grows with β from 1.3 % at $\beta = 0$ to 4.0 % at $\beta = \pi/2$, with a corresponding phase error of at most 7 mrad. The Re/Im-resolved breakdown (Tables 7, 8) reveals substantially stronger anisotropy: $\operatorname{Im} S_{\text{fw},\theta}$ at $\beta = \pi/2$ reaches 10.5 %, reflecting the concentration of the plasmonic surface mode in the sub-wavelength inter-sphere gap when the doublet axis is aligned with the incident polarisation.

Table 6 reports the composite $|S_{\text{fw,mean}}|$ magnitudes and complex relative errors, while Tables 7 and 8 break the complex amplitudes into their real and imaginary parts for the θ and φ channels respectively. The relative errors are defined pointwise as

$$\text{rel. err.}^{\text{Re}} = \frac{\left|\operatorname{Re} S^{\text{VIEM}}_{\text{fw},\theta} - \operatorname{Re} S^{\text{MSTM}}_{\text{fw},\theta}\right|}{\left|\operatorname{Re} S^{\text{MSTM}}_{\text{fw},\theta}\right|},$$

$$\text{rel. err.}^{\text{Im}} = \frac{\left|\operatorname{Im} S^{\text{VIEM}}_{\text{fw},\theta} - \operatorname{Im} S^{\text{MSTM}}_{\text{fw},\theta}\right|}{\left|\operatorname{Im} S^{\text{MSTM}}_{\text{fw},\theta}\right|},$$

and analogously for $S_{\text{fw},\varphi}$.

Table 6: Summary of the complex $S_{\text{fw,mean}}$ agreement at the fully-converged $N_{\text{trunc}} = 15$. Geometry as in the text: equal spheres of radius 0.030 µm, gap 0.003 µm, $\lambda_0 = 0.638$ µm, $m_m = 1$.

material	β [rad]	$\lvert S^{\text{VIEM}}_{\text{fw,mean}}\rvert$	$\lvert S^{\text{MSTM}}_{\text{fw,mean}}\rvert$	rel. $\lvert\cdot\rvert$ err.	complex rel. err.	phase err. [rad]
dielectric	0	1.769×10^{-3}	1.794×10^{-3}	1.4 %	1.4 %	1.3×10^{-4}
$m_p = 1.60 + 0.01i$	$\pi/4$	1.822×10^{-3}	1.849×10^{-3}	1.5 %	1.5 %	1.6×10^{-4}
	$\pi/2$	1.878×10^{-3}	1.908×10^{-3}	1.6 %	1.6 %	2.0×10^{-4}
gold	0	6.420×10^{-3}	6.504×10^{-3}	1.3 %	1.3 %	5.1×10^{-4}
$m_p = 0.175 + 3.48i$	$\pi/4$	8.043×10^{-3}	8.287×10^{-3}	2.9 %	3.0 %	4.9×10^{-3}
	$\pi/2$	9.785×10^{-3}	1.020×10^{-2}	4.0 %	4.1 %	7.0×10^{-3}

Discussion. For the high-index dielectric doublet the agreement is essentially polarization- and orientation-independent: $\operatorname{Re} S_{\text{fw},\theta}$ and $\operatorname{Re} S_{\text{fw},\varphi}$ are both recovered to 1.4 %–1.7 %, and the imaginary parts to ~ 2 %–3 % (Tables 7, 8). The extra factor on the imaginary axis is a purely scale effect: $|\operatorname{Im} S_{\text{fw,mean}}|$ is 30–50 times smaller than $|\operatorname{Re} S_{\text{fw,mean}}|$ in the weakly absorbing

Table 7: θ-channel forward amplitude $S_{\mathrm{fw},\theta}$: real and imaginary parts from block-VIEM.jl and MSTMforCAS.jl, and the corresponding relative errors.

material	β [rad]	$\mathrm{Re}\, S_\theta^{\mathrm{VIEM}}$	$\mathrm{Re}\, S_\theta^{\mathrm{MSTM}}$	rel. err. Re	$\mathrm{Im}\, S_\theta^{\mathrm{VIEM}}$	$\mathrm{Im}\, S_\theta^{\mathrm{MSTM}}$	rel. err. Im
dielectric	0	1.7682×10^{-3}	1.7934×10^{-3}	1.4 %	4.1063×10^{-5}	4.1991×10^{-5}	2.2 %
	$\pi/4$	1.8768×10^{-3}	1.9066×10^{-3}	1.6 %	4.7872×10^{-5}	4.9082×10^{-5}	2.5 %
	$\pi/2$	1.9933×10^{-3}	2.0279×10^{-3}	1.7 %	5.5225×10^{-5}	5.6771×10^{-5}	2.7 %
gold	0	6.4046×10^{-3}	6.4886×10^{-3}	1.3 %	4.3635×10^{-4}	4.4705×10^{-4}	2.4 %
	$\pi/4$	9.6439×10^{-3}	1.0037×10^{-2}	3.9 %	1.2147×10^{-3}	1.3381×10^{-3}	9.2 %
	$\pi/2$	1.3102×10^{-2}	1.3818×10^{-2}	5.2 %	2.0467×10^{-3}	2.2857×10^{-3}	10.5 %

Table 8: φ-channel forward amplitude $S_{\mathrm{fw},\varphi}$: real and imaginary parts from block-VIEM.jl and MSTMforCAS.jl, and the corresponding relative errors.

material	β [rad]	$\mathrm{Re}\, S_\varphi^{\mathrm{VIEM}}$	$\mathrm{Re}\, S_\varphi^{\mathrm{MSTM}}$	rel. err. Re	$\mathrm{Im}\, S_\varphi^{\mathrm{VIEM}}$	$\mathrm{Im}\, S_\varphi^{\mathrm{MSTM}}$	rel. err. Im
dielectric	0	1.7683×10^{-3}	1.7934×10^{-3}	1.4 %	4.1285×10^{-5}	4.1991×10^{-5}	1.7 %
	$\pi/4$	1.7651×10^{-3}	1.7902×10^{-3}	1.4 %	4.2005×10^{-5}	4.2724×10^{-5}	1.7 %
	$\pi/2$	1.7619×10^{-3}	1.7870×10^{-3}	1.4 %	4.2680×10^{-5}	4.3463×10^{-5}	1.8 %
gold	0	6.4049×10^{-3}	6.4886×10^{-3}	1.3 %	4.3967×10^{-4}	4.4705×10^{-4}	1.7 %
	$\pi/4$	6.3556×10^{-3}	6.4392×10^{-3}	1.3 %	4.4777×10^{-4}	4.5502×10^{-4}	1.6 %
	$\pi/2$	6.3084×10^{-3}	6.3893×10^{-3}	1.3 %	4.5536×10^{-4}	4.6332×10^{-4}	1.7 %

case, so a fixed absolute discretisation error appears larger on Im in relative terms. Both parts converge as $O(h^2)$ under mesh refinement (Sec. 8.1).

The plasmonic gold doublet reveals a strong *orientation-polarization anisotropy* of the discretisation error. The φ channel (Table 8) is essentially flat in β at $\approx 1.3\,\%$ on Re and 1.6 %–1.7 % on Im, because $S_{\mathrm{fw},\varphi}$ probes the polarisation component transverse to both the incidence direction and the doublet axis – the two spheres are essentially independent through this channel. The θ channel (Table 7) behaves very differently: it couples to the component of the incident electric field that lies in the plane spanned by $\hat{\boldsymbol{k}}_{\mathrm{inc}}$ and the doublet axis, and at $\beta = \pi/2$ this component is aligned with the doublet axis. The two gold monomers are then driven in phase along their centre-to-centre line, with the interaction field concentrated in the sub-wavelength inter-sphere gap where the plasmonic surface mode lives. In this configuration the error rises to 5.2 % on $\mathrm{Re}\, S_{\mathrm{fw},\theta}$ and 10.5 % on $\mathrm{Im}\, S_{\mathrm{fw},\theta}$.

Verified mesh-refinement convergence. To confirm that the residual error is a bulk discretisation error (and not, for instance, a skin-layer-resolution plateau), four Phase A refinement runs were made at $\ell_c = R/k$ for $k = 5, 6, 7, 8$ and compared against the same MSTM $N = 15$ reference (raw data `benchmarks/cas_v2/doublet_mstm/phase_a_memory.json`, error-table generator `compute_refinement_errors.py`). The log-log fit

$$\text{rel. err.}(\ell_c) \;\propto\; (\ell_c/R)^p \tag{58}$$

is evaluated for the five CAS-v2 components at $\beta = \pi/2$:

component	relative error at $\ell_c = R/k$				fitted p
	$k=5$	$k=6$	$k=7$	$k=8$	(4-pt)
$\lvert S_{\mathrm{fw,mean}}\rvert$	4.36 %	2.73 %	1.94 %	1.60 %	2.16
$\mathrm{Re}\, S_{\mathrm{fw},\theta}$	5.61 %	3.47 %	2.44 %	2.05 %	2.17
$\mathrm{Im}\, S_{\mathrm{fw},\theta}$	10.72 %	6.43 %	4.62 %	4.07 %	2.10
$\mathrm{Re}\, S_{\mathrm{fw},\varphi}$	1.38 %	0.97 %	0.73 %	0.50 %	2.11
$\mathrm{Im}\, S_{\mathrm{fw},\varphi}$	2.03 %	1.82 %	1.38 %	0.58 %	2.49

The theoretical linear-SWG interior rate is $p = 2$, and the 4-point fit recovers it cleanly across every component: all five exponents lie in $[2.10, 2.49]$, with the stiffest observable $\mathrm{Im}\, S_{\mathrm{fw},\theta}$ at $p = 2.10$ — essentially the theoretical rate. An earlier two-point fit using only $R/5$ and $R/6$ had given $p \in [0.78, 3.39]$ (with $\mathrm{Im}\, S_{\mathrm{fw},\theta}$ the *fastest* at $p = 3.39$); that spread is now understood as a two-point artefact. The clean $p \approx 2$ convergence holding uniformly across θ and φ channels confirms that the skin-layer resolution is not a rate-limiting factor once $\ell_c \leq R/5$.

Phase A (Sec. 5.5) has reduced the AIM memory footprint to a near-linear $\mathcal{O}(N_{\mathrm{dof}}^{1.02})$ scaling (Table 1), so the $R/7$ and $R/8$ rows above fit in 263 MiB and 413 MiB of operator memory respectively. Extrapolating the measured $p = 2.10$ slope of $\mathrm{Im}\, S_{\mathrm{fw},\theta}$ predicts $4.07\,\% \times (8/10)^{2.10} \approx 2.6\,\%$ at $\ell_c = R/10$, and the $|S_{\mathrm{fw,mean}}|$ error $1.60\,\% \times (8/10)^{2.16} \approx 1.0\,\%$; the projected 760 MiB operator memory at $R/10$ is feasible on a 16 GB workstation, and that run is the next planned step.

Physically, the gold skin depth at 638 nm is

$$\delta = \frac{\lambda_0}{2\pi\, \mathrm{Im}\, m_p} \approx 29\,\mathrm{nm},$$

essentially equal to the monomer radius R itself. With the present $\ell_c = R/5 \approx 6\,\mathrm{nm}$ mesh, δ is resolved by roughly five linear tetrahedra – adequate for few-percent accuracy, but the imaginary part of the axis-aligned-polarisation channel becomes the limiting observable. Refining to $\ell_c = R/8 - R/10$ (or enabling the boundary-face half-SWG correction with higher-order Duffy rules) is expected to bring the $\mathrm{Im}\, S_{\mathrm{fw},\theta}$ error below 3 % uniformly, since the SWG basis converges as $O(h^2)$ under uniform refinement.

The practical takeaway is that for high-$|m|$ absorbing aggregates the imaginary part of the longitudinal-polarisation channel is the stiffest observable *in absolute terms* and should be adopted as the convergence indicator when sizing ℓ_c: a 4.0 % error on $|S_{\mathrm{fw,mean}}|$ at $\beta = \pi/2$ *hides* a 10.5 % error on $\mathrm{Im}\, S_{\mathrm{fw},\theta}$, and it is the Re/Im-resolved component view of Tables 7–8 that makes this gap visible. The good news is that the component also converges the fastest under refinement ($p \approx 3.4$ measured), so modest mesh refinement is highly cost-effective here.

A second, orthogonal lesson concerns the adequacy of the exact reference itself. MSTM-forCAS's default auto-truncation delivers $N_{\mathrm{trunc}} = 3$ at these parameters – adequate for the dielectric case, where $|a_3|$ is already below 10^{-6} of $|a_1|$, but badly *under-truncating* the gold doublet: $|S_{\mathrm{fw,mean}}|$ shifts by $\approx 5\,\%$ between $N = 3$ and the fully-converged $N = 15$ reference. We therefore use $N_{\mathrm{trunc}} = 15$, which reaches rtol $\leq 10^{-6}$ on both materials at $x \approx 0.3$. Before the three MSTMforCAS stability fixes listed above this convergence sweep was impossible – the package could not take N beyond 6 at $x \approx 0.3$ because of combined Riccati–Bessel and translation-coefficient numerical instabilities. For aggregates of plasmonic monomers the appropriate MSTM truncation is set by the *inter-sphere coupling* amplified by $|m \cdot x|$, not by the single-sphere Mie convergence, and MSTM reference data should always be supplied with an explicit N_{trunc} and a convergence check (`benchmarks/cas_v2/doublet_mstm/check_trunc_convergence.jl`).

The full benchmark pipeline (geometry builder, VIEM solve, MSTM reference, relative-error report) lives under `benchmarks/cas_v2/doublet_mstm/` and uses a shared `config.jl` so the two codes solve the exact same physical problem.

9 Spheroid Sweep HDF5 Output

For parameter-sweep applications the forward scattering amplitudes on a five-dimensional grid are written to an HDF5 file whose layout is deliberately identical to that produced by `block-DDA_Py` [11], so the downstream consumer `PCAS_Bayes_APM_Nonspherical/lut_generation/build_spheroid_lut.py` reads VIEM output without modification. The schema is documented in `block-DDA_Py/.claude/spheroid_h5_schema_spec.md` and summarised here for completeness.

9.1 Parameter grid

The five grid axes are stored at the root level of the file and shared across wavelength groups:

- `D_ve_grid` — volume-equivalent diameter D_{ve} [µm];
- `RI_real_grid` — real part of the particle refractive index, $\mathrm{Re}(m_p)$;
- `log_AR_grid` — base-10 logarithm of bc_ratio $= b/c$, following the `block-DDA_Py` convention where bc_ratio > 1 is oblate, < 1 is prolate, and $= 1$ is a sphere;
- `cos_theta_o_half_grid` — $\cos\theta_o$ on $[0, 1]$ (half-domain, extended to $[-1, 1]$ by the consumer using the $\theta_o \to \pi - \theta_o$ mirror symmetry);
- `phi_o_grid` — ϕ_o on $[0, \pi]$ (half-domain, extended to $[0, 2\pi]$ by the consumer using the ϕ_o-periodicity $S(\theta_o, \phi_o + \pi) = S(\theta_o, \phi_o)$).

All axes must be equidistant; `write_spheroid_sweep_h5` enforces this constraint and raises an error otherwise, because the downstream `NdimSpline_JAX` consumer assumes uniform grids.

9.2 Per-wavelength datasets

Results for each wavelength λ_0 are stored under a group `wl_<value>` (e.g. `wl_0p638` for $\lambda_0 = 0.638\,\mu\mathrm{m}$), with the attributes `wl_0` and `m_m`. Each group contains

- `S_fw_theta_re`, `S_fw_theta_im`, `S_fw_phi_re`, `S_fw_phi_im`: the real and imaginary parts of $S_{\mathrm{fw},\theta}$ and $S_{\mathrm{fw},\phi}$ on the shape $(N_{D_{ve}}, N_{RI}, N_{AR}, N_{u_{1/2}}, N_{\phi_o})$;
- `converged`: a boolean array of shape $(N_{D_{ve}}, N_{RI}, N_{AR})$ indicating successful solves. Non-converged cells carry `NaN+NaN`i in the four amplitude arrays across all $(N_{u_{1/2}}, N_{\phi_o})$ entries.

The ϕ_o axis corresponds exactly to the Euler angle α, and the populating code applies the analytical α-expansion (52)–(53) to the $\alpha = 0$ solves, so only $N_{u_{1/2}}$ independent VIEM solves are required per $(D_{ve}, \mathrm{Re}(m_p), \mathrm{AR})$ triple.

10 AIM-accelerated block-Krylov multi-orientation solve

For large orientation sweeps — orientation LUTs typically require $L = 10^2$–10^3 incidence directions per $(D_{ve}, \mathrm{Re}(m_p), \mathrm{AR})$ cell — reusing the same LU factorization of the dense impedance matrix (Sec. **??** style) becomes infeasible as soon as $N \gg 10^3$. For those problems `block-VIEM.jl` v0.2.0 exposes an AIM-accelerated *block-Krylov* path that solves

$$\mathbf{Z}\,\mathbf{X} = \mathbf{B}, \qquad \mathbf{X}, \mathbf{B} \in \mathbb{C}^{N \times L}, \tag{59}$$

with a single pre-built AIM operator and a single block iteration that acts on all L right-hand sides simultaneously.

Block MVP. The AIM operator supports the multi-RHS form

$$\mathrm{aim\_mvp}(\mathbf{X}) \;=\; \tfrac{1}{\epsilon_p}\mathbf{M}\mathbf{X} \;-\; \tfrac{\kappa}{\epsilon_{\mathrm{bg}}}\big[\mathbf{K}_{\mathrm{AIM}}(\mathbf{X}) + \mathbf{K}_{\mathrm{near}}\mathbf{X}\big] \;+\; \mathbf{K}_{\text{half-SWG}}\mathbf{X},$$

where $\mathbf{M}, \mathbf{K}_{\mathrm{near}}, \mathbf{K}_{\text{half-SWG}}$ are sparse and their matrix–matrix products cost $\mathcal{O}(\mathrm{nnz} \cdot L)$ in a single BLAS call. The far-field convolution $\mathbf{K}_{\mathrm{AIM}}(\mathbf{X})$ is applied column-by-column and the L columns are dispatched across the available Julia threads via `Threads.@spawn`; the FFT plans themselves are shared and use the process-wide FFTW thread pool, so the three levels of parallelism — column, FFT, and iteration — can be balanced as $n_{\mathrm{col}} \times n_{\mathrm{FFT}} \approx n_{\mathrm{cores}}$ via `BlockVIEM.set_fft_threads`. For a single 20-core machine at $L = 8$ we measure a 2.6× per-column speed-up over the serial baseline.

Block BiCGSTAB. The default solver follows Tadano, Sakurai and Kuramashi [13] and is a direct port of `block-DDA_Py`'s `bl_bicgstab_jacobi_mvp_fft`, sans the Jacobi preconditioner (which is cheap for DDA because the dipole self-term is diagonal, but would require $\mathcal{O}(N)$ FFT probes for VIEM; the SWG mass matrix already normalizes the system adequately). Given $\mathbf{B} \in \mathbb{C}^{N\times L}$, set $\mathbf{X}_0 = \mathbf{0}$, $\mathbf{R}_0 = \mathbf{B}$, $\mathbf{P} = \mathbf{R}_0$, and $\widetilde{\mathbf{R}}_0 = \mathbf{R}_0$, then iterate

$$\mathbf{V}_k = \mathbf{A}\mathbf{P}_k, \tag{60}$$

$$\boldsymbol{\alpha}_k = \big(\widetilde{\mathbf{R}}_0^H\mathbf{V}_k\big)^{-1}\big(\widetilde{\mathbf{R}}_0^H\mathbf{R}_k\big), \tag{61}$$

$$\mathbf{T}_k = \mathbf{R}_k - \mathbf{V}_k\boldsymbol{\alpha}_k, \tag{62}$$

$$\mathbf{Z}_k = \mathbf{A}\mathbf{T}_k, \tag{63}$$

$$\xi_k = \frac{\mathrm{tr}\big(\mathbf{Z}_k^H\mathbf{T}_k\big)}{\mathrm{tr}\big(\mathbf{Z}_k^H\mathbf{Z}_k\big)}, \tag{64}$$

$$\mathbf{X}_{k+1} = \mathbf{X}_k + \mathbf{P}_k\boldsymbol{\alpha}_k + \xi_k\mathbf{T}_k, \tag{65}$$

$$\mathbf{R}_{k+1} = \mathbf{T}_k - \xi_k\mathbf{Z}_k, \tag{66}$$

$$\boldsymbol{\beta}_k = -\big(\widetilde{\mathbf{R}}_0^H\mathbf{V}_k\big)^{-1}\big(\widetilde{\mathbf{R}}_0^H\mathbf{Z}_k\big), \tag{67}$$

$$\mathbf{P}_{k+1} = \mathbf{R}_{k+1} + \big(\mathbf{P}_k - \xi_k\mathbf{V}_k\big)\boldsymbol{\beta}_k, \tag{68}$$

stopping once $\|\mathbf{R}_{k+1}\|_F/\|\mathbf{B}\|_F <$ tol. Each outer iteration performs exactly two block MVPs ($\mathbf{AP}$ and $\mathbf{AT}$) and four $L \times L$ Gram matrices, so the per-iteration cost is $\mathcal{O}\big(T_{\mathrm{MVP}} \cdot L + NL^2\big)$. Block BiCGSTAB can break down when the columns of $\mathbf{R}_k$ lose numerical independence; in practice this is rare for the physically distinct incidence directions of a CAS-v2 LUT, but the fallback below exists.

Block GMRES. The fallback is unrestarted Block GMRES [14]: a block-Arnoldi recurrence with thin QR factorization of each new block Krylov vector builds an orthonormal basis $\mathbf{V}_1, \mathbf{V}_2, \ldots$ of the block Krylov space $\mathcal{K}_k(\mathbf{A}; \mathbf{B})$, together with an upper block-Hessenberg matrix $\underline{\mathbf{H}}_k$. At iteration k the approximate solution is $\mathbf{X}_k = \mathbf{X}_0 + [\mathbf{V}_1|\cdots|\mathbf{V}_k]\mathbf{Y}_k$, where $\mathbf{Y}_k$ minimizes $\|\underline{\mathbf{H}}_k\mathbf{Y} - \mathbf{E}_1\boldsymbol{\Lambda}\|_F$ with $\boldsymbol{\Lambda}$ the thin-QR factor of the initial residual. Unlike BiCGSTAB the memory grows linearly in the iteration count (k blocks of shape $N \times L$ are retained), so in practice Block GMRES is used as a robust fallback rather than a default.

Numerical verification. On an oblate AR$=3$ spheroid ($D_{ve} = 0.40\,\mu\mathrm{m}$, $m_p = 1.5$, $\lambda_0 = 0.638\,\mu\mathrm{m}$, $N = 2009$ half-SWG DoFs) the two block solvers reach tolerance 10^{-8} in roughly 50 block iterations on $L = 5$ orientations, and their CAS-v2 forward amplitudes agree to $\sim 10^{-10}$ relative. Compared to the dense LU reference, the AIM block-Krylov observables agree to $\sim 0.8\%$, which is the AIM discretization error at the chosen pitch and not a Krylov convergence issue. The analytical $+2j\alpha$ spheroid symmetry (Sec. **??**) on the AIM block-BiCGSTAB solution holds at $\sim 5 \times 10^{-10}$, confirming that the block iteration preserves orientation symmetry to the solver tolerance.

11 Gaussian Random Ellipsoid (GRE) Shape Model

Block-DDA_Py parameterises particle geometry via the Gaussian Random Ellipsoid (GRE) shape model [16]. `block-VIEM.jl` v0.3.0 provides a Julia-native implementation (`src/gre_mesh.jl`) that accepts the same four-parameter tuple ($r_{v,\mathrm{base}}$, b/c, a/b, β) and produces a tetrahedral mesh suitable for the VIEM solver.

11.1 Base ellipsoid

The semi-axes of the undeformed ellipsoid are determined by the volume-preservation condition $V = \frac{4\pi}{3}abc = \frac{4\pi}{3}r_{v,\text{base}}^3$:

$$c = \Big(\frac{r_{v,\text{base}}^3}{(a/b)(b/c)^2}\Big)^{1/3}, \qquad b = c\,(b/c), \qquad a = b\,(a/b). \tag{69}$$

The convention is $a \geq b \geq c$, with c along the z-axis.

11.2 Gaussian random deformation field

A zero-mean Gaussian random field $s(\theta, \phi)$ is sampled on the ellipsoid surface with covariance

$$C(\boldsymbol{r}_i, \boldsymbol{r}_j) = \beta^2 \exp\Big(-\frac{\|\boldsymbol{r}_i - \boldsymbol{r}_j\|^2}{2\,\ell_c^2}\Big), \qquad \ell_c = 0.3\,c, \tag{70}$$

where $\{\boldsymbol{r}_i\}$ are the ellipsoid-surface positions on a coarse (θ, ϕ) grid ($N_\theta = 25$, $N_\phi = 100$). Sampling is performed via Cholesky decomposition $\mathbf{C} = \mathbf{L}\mathbf{L}^\top$, $\boldsymbol{s} = \mathbf{L}\boldsymbol{z}$, $\boldsymbol{z} \sim \mathcal{N}(0, \mathbf{I})$. The coarse-grid field is bilinearly interpolated onto a finer grid (interpolation factor $\max\big(1, \lfloor 4\,(bc\_ratio \cdot ab\_ratio)^{1/3}\rceil\big)$).

Each surface point is then displaced along the outward ellipsoid normal $\hat{\boldsymbol{n}}$:

$$\boldsymbol{r}_{\text{GRE}} = \boldsymbol{r}_0 + h_0\big(e^{s(\theta,\phi)} - \tfrac{1}{2}\beta^2 - 1\big)\,\hat{\boldsymbol{n}}, \qquad h_0 = \frac{c^2}{a}, \tag{71}$$

reproducing the `compute_r_points_on_GRE` definition in block-DDA_Py exactly.

11.3 Mesh generation strategy

$\beta = 0$ (smooth ellipsoid). The Gmsh OpenCASCADE kernel constructs the ellipsoid analytically (`addSphere` + `dilate`), yielding a high-quality mesh without intermediate geometry conversion.

$\beta > 0$ (deformed GRE).

1. Construct the base ellipsoid via OCC and generate the full 3-D tetrahedral mesh at the target characteristic length ℓ_c.
2. Deform every mesh node (surface *and* interior) according to the sampled field (71). For an interior node at position $\boldsymbol{r}$ with normalised ellipsoidal radius $\tilde{r} = \sqrt{(x/a)^2 + (y/b)^2 + (z/c)^2} \in [0, 1]$, the displacement is scaled linearly:

$$\boldsymbol{r}_{\text{new}} = \boldsymbol{r} + \tilde{r}\,\delta(\theta, \phi)\,\hat{\boldsymbol{n}}, \tag{72}$$

where $\delta = h_0(e^s - \frac{1}{2}\beta^2 - 1)$. This ensures zero displacement at the centre and full displacement at the surface, preserving element quality for moderate β.

This two-step approach avoids the fragile STL $\to$ `classifySurfaces` $\to$ `createGeometry` pipeline in Gmsh, which fails for closed surfaces without boundary edges.

11.4 Adaptive mesh size

The characteristic edge length ℓ_c is set automatically as the minimum of three constraints:

Constraint	Formula	Condition
Wavelength	$\lambda_0/(\max\lvert m_p\rvert \cdot N_{pw})$	λ_0 and m_p supplied
Geometry	$c/3$	always
Correlation length	$\ell_c^{\text{corr}}/3 = 0.1\,c$	$\beta > 0$ only

Block-DDA_Py uses $N_{pw} = \text{dpl} = 17$ dipoles per wavelength inside the particle. The higher per-DOF accuracy of SWG allows $N_{pw} \approx 10$ for VIEM, corresponding to comparable overall accuracy at fewer degrees of freedom.

11.5 Volume-equivalent radius

After meshing, the volume-equivalent radius is computed directly from the tetrahedral mesh:

$$V_{\text{mesh}} = \sum_i V_i, \qquad r_{ve} = \left(\frac{3\,V_{\text{mesh}}}{4\pi}\right)^{1/3}. \tag{73}$$

Unlike the DDA approach (which rescales the lattice spacing to enforce $V = \frac{4\pi}{3} r_{v,\text{base}}^3$ exactly), the VIEM mesh volume is geometrically accurate to the mesh resolution and no post-hoc rescaling is needed.

11.6 Cross-validation vs block-DDA_Py ($\beta > 0$)

To verify that the GRE mesh generation and VIEM solver produce correct observables for deformed particle shapes, we cross-validate against block-DDA_Py on three GRE configurations with $\beta > 0$. The *same* Gaussian random deformation field $s(\theta, \phi)$ (generated and saved by the Python script `benchmarks/cas_v2/gre_comparison/run_dda_gre.py`) is loaded by the Julia script (`run_viem_gre.jl`) and used to deform the VIEM mesh via `gre_mesh_with_field`, ensuring that both codes solve the identical shape.

Common parameters: $\lambda_0 = 0.638\,\mu\text{m}$, $n_m = 1.0$, $m_p = 1.5 + 0.01\,i$, $r_{v,\text{base}} = 0.20\,\mu\text{m}$. The DDA uses dpl $= 17$ ($N_{\text{DDA}} \approx 2300$ dipoles); the VIEM uses $\ell_c = 0.035\,\mu\text{m}$ ($N_{\text{VIEM}} \approx 8600$ half-SWG DoFs). Two tilt angles $\beta_{\text{ori}} = \pi/4$ and $\pi/2$ are tested.

Case	b/c	a/b	β	$\lvert\Delta S_\theta\rvert/\lvert S_\theta\rvert$	$\lvert\Delta S_\phi\rvert/\lvert S_\phi\rvert$
A	2.0	1.0	0.10	0.5–0.7 %	0.1–0.3 %
B	3.0	1.0	0.15	2.0–2.6 %	1.7–1.8 %
C	1.5	1.5	0.20	1.2–1.6 %	1.6–1.8 %

The relative errors in the complex forward amplitudes S_θ and S_ϕ range from 0.1% to 2.6%, consistent with the discretisation-error level observed in the smooth-spheroid benchmark (Sec. 11). The errors are larger for higher β and b/c, reflecting both the coarser per-feature resolution of the DDA lattice on thin or strongly deformed shapes and the inherent discretisation mismatch between cubic-lattice DDA and tetrahedral VIEM.

12 Anisotropic Permittivity

Version 0.4.0 extends the EFVIE-D formulation to diagonal-tensor permittivity $\bar{\epsilon}_p = \text{diag}(\epsilon_x, \epsilon_y, \epsilon_z)$ in the particle frame, matching the `m_p_xyz` parameter of block-DDA_Py.

12.1 Modified weak form

Defining the component-wise contrast $\kappa_\alpha = (\epsilon_\alpha - \epsilon_{\text{bg}})/\epsilon_\alpha$ and its average $\bar{\kappa} = (\kappa_x + \kappa_y + \kappa_z)/3$, the Galerkin impedance element becomes

$$Z_{mn} = \sum_\alpha \frac{1}{\epsilon_\alpha} \int (f_m)_\alpha \, (f_n)_\alpha \, dV - \frac{1}{\epsilon_{\text{bg}}} \iint \Big[k_0^2 \sum_\alpha \kappa_\alpha \, (f_m)_\alpha \, (f_n)_\alpha \; - \; \bar{\kappa} \, (\nabla \cdot \boldsymbol{f}_m) \, (\nabla' \cdot \boldsymbol{f}_n) \Big] G \, dV' \, dV. \tag{74}$$

The simplification $\nabla' \cdot (\bar{\kappa} \cdot \boldsymbol{f}_n) = \bar{\kappa} \, \nabla' \cdot \boldsymbol{f}_n$ exploits the SWG structure $\partial (f_n)_\alpha / \partial r_\alpha = a_n/(3V)$ (identical for all α).

12.2 Implementation

- **Mass term:** component-weighted inner product $\sum_\alpha (1/\epsilon_\alpha) \int (f_m)_\alpha (f_n)_\alpha \, dV$.
- **Radiation kernel (bulk):** k_0^2 term uses per-component κ_α; $\nabla\nabla\cdot$ term uses $\bar{\kappa}$.
- **Half-SWG surface terms** (K^B, K^C, K^D): scaled by $\bar{\kappa}$ (exact for isotropic; $O(|\Delta\kappa|/|\kappa|)$ approximation for anisotropic).
- **AIM FFT-MVP:** per-channel κ_α weighting on the x, y, z vector channels; $\bar{\kappa}$ on the divergence channel.
- **Far-field amplitude:** $\boldsymbol{F} = (k_0^2/\epsilon_{\text{bg}})(\mathbf{I} - \hat{\boldsymbol{r}}\hat{\boldsymbol{r}}) \cdot (\bar{\kappa} \cdot \boldsymbol{P})$.
- **Absorption:** $C_{\text{abs}} = \frac{k_0}{\epsilon_{\text{bg}}} \sum_\alpha \frac{\text{Im}(\epsilon_\alpha)}{|\epsilon_\alpha|^2} \int |D_\alpha|^2 \, dV$.

When $\epsilon_x = \epsilon_y = \epsilon_z$ all expressions reduce to the isotropic formulation of Sec. **??** and earlier sections.

12.3 Cross-validation vs block-DDA_Py

A sphere ($r_{ve} = 0.20\,\mu\text{m}$, $\lambda_0 = 0.638\,\mu\text{m}$, $n_m = 1$, $\ell_c = 0.035\,\mu\text{m}$, $N_{\text{VIEM}} \approx 8000$ half-SWG DoFs) was solved with three refractive-index configurations at tilt angles $\beta_{\text{ori}} = \pi/4$ and $\pi/2$:

Case	$m_p = [m_x,\ m_y,\ m_z]$	$\lvert\Delta S_\theta\rvert/\lvert S_\theta\rvert$	$\lvert\Delta S_\phi\rvert/\lvert S_\phi\rvert$
iso	[1.5, 1.5, 1.5]	1.3–2.4 %	2.4 %
mild	[1.55, 1.5, 1.45]	1.2–3.5 %	2.3–2.6 %
strong	[1.6, 1.5, 1.4]	0.7–5.0 %	2.3–2.9 %

The isotropic baseline (1.3–2.4 %) matches the existing spheroid benchmark accuracy. Errors grow moderately with anisotropy strength, reaching $\sim 5\%$ for the "strong" case at $\beta = \pi/2$, which is dominated by the $\bar{\kappa}$ approximation in the half-SWG surface terms and the cubic-vs-tetrahedral discretisation mismatch.

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